Online Threshold Quantum Search for a Reproducible Low-Thrust Cislunar Missed-Thrust Benchmark

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Abstract

This paper introduces a quantum-specific, constraint-preserving search experiment for a reproducible normalized Earth–Moon CR3BP low-thrust missed-thrust initialization benchmark. The primary discrete problem is a cardinality benchmark: among feasible high-duty thrust schedules, choose low-error schedules while respecting a fixed schedule family. The headline method is an online threshold-amplitude-amplification simulation inspired by BBHT and Durr–Hoyer minimum finding; it starts from a sampled incumbent, marks only states below the current incumbent threshold, samples from the amplified distribution, and updates the incumbent only after an evaluated improvement. It does not mark a known top-2 set in advance. On the recorded branch-refined 66-state `k10_full` family, the online protocol reaches global-optimum hit rates 0.235/0.62/0.935 at $\lceil\sqrt{M}\rceil$, $2\lceil\sqrt{M}\rceil$, and $4\lceil\sqrt{M}\rceil$ total-oracle budgets, versus 0.07/0.20/0.58 for uniform random sampling without replacement and 0.10–0.625 for matched-budget finite-subspace simulated annealing, cross-entropy, and genetic baselines. Larger CR3BP feedback-propagation screening cases extend the query-model scale to $M = 1820$ and $M = 4845$, but they are front-end screening results, not branch-refined feasible trajectory evidence; their targeted branch-refinement diagnostic still fails the nominal threshold. The classical baselines also show that the online protocol is not universally dominant: the genetic baseline reaches 0.935 on the $M = 4845$ screening case at $4\lceil\sqrt{M}\rceil$. Ranked screening is an optimistic ceiling/diagnostic, and the earlier top-2 Grover protocol is retained only as a diagnostic ablation. The result is a bounded query-model initializer benchmark: it is not hardware evidence, a quantum-advantage claim, high-fidelity validation, or a replacement for continuous trajectory optimization. The practical value is testing a threshold-only quantum search as a complementary constrained-subspace initializer for candidate discovery

under fixed evaluation budgets.

Keywords: Trajectory optimization benchmark, low-thrust trajectory optimization, cislunar dynamics, missed-thrust recovery, robust initialization, online threshold search, amplitude amplification, CR3BP

1. Introduction

Electric low-thrust propulsion enables propellant-efficient cislunar transfers, but it also makes trajectory design highly sensitive to the initial guess supplied to a nonlinear optimizer. A practical design must determine both continuous thrust magnitudes and directions and a qualitative maneuver structure: which intervals thrust, coast, or recover after an interruption. For astrodynamics researchers, this creates a reproducibility problem as much as an algorithm problem. A proposed initializer is difficult to interpret unless the schedule objective, missed-thrust masks, continuous recovery backend, evaluation budget, and failure cases are all visible.

Missed-thrust robustness is a demanding test of that pipeline. Safe modes, propulsion interruptions, or operations-driven thrust outages can invalidate a nominal low-thrust transfer. Grebow and McCarty show that including missed-thrust branches in low-thrust cislunar design can greatly reduce recovery propellant for week-scale outages [1]. Sinha and Beeson identify initial-guess generation as a bottleneck for low-thrust trajectory design with missed-thrust robustness [2]. This paper therefore asks a constrained quantum-search question inside a benchmark setting:

> If cardinality-constrained missed-thrust schedules form the feasible subspace, can an online threshold-amplitude-amplification protocol, which updates its incumbent by evaluated schedule costs rather than by a predeclared top-k oracle, find low-cost schedules with fewer explicit candidate evaluations than uniform random sampling under the same total oracle/evaluation budget?

Binary and QUBO formulations are included because they are a natural way to encode thrust-window availability, and because recent low-thrust studies have explored binary/QUBO and quantum-assisted formulations [3, 4]. The new contribution is narrower and more quantum-specific than the earlier generic bitstring story: it treats cardinality-constrained schedules as the Hilbert-space basis and applies an online threshold feasible-subspace search rather than a penalty over infeasible bitstrings or a predeclared top-k marked set. The continuous trajectory remains a classical optimal-control problem; the quantum layer only proposes discrete schedules for branch-recovery refinement or, in the larger screening tier, for CR3BP feedback-propagation ranking.

The paper makes four contributions for a trajectory-optimization audience. First, it defines and archives a normalized Earth–Moon CR3BP benchmark resource with source states, missed-thrust masks, configurations, result artifacts, generated tables/figures, metadata, and a machine-readable manifest. Second, it adds an online threshold-amplitude-amplification simulation for finite feasible schedule subspaces. The implemented oracle marks only states with cost below the current incumbent; known top-2 schedules are not supplied to the headline protocol. Third, it archives the earlier finite threshold/top-k Grover sweep as an exploratory diagnostic and reports baselines with explicit semantics: uniform random sampling, finite-subspace simulated annealing, cross-entropy, and genetic search are deployable matched-budget classical comparators, while ranked screening is a ceiling/diagnostic for what a near-perfect classical ranking signal would do, not a deployable baseline that the online protocol is claimed to beat. Existing structured-QAOA and generic `qaoa_statevector` rows remain auxiliary diagnostics with no hardware or advantage claim. Fourth, it keeps the continuous-backend and replay/provenance ladder that separates selected-branch evidence, all-mask diagnostics, all-configured-mask evidence, accepted-control replay, and recorded force-model stress.

The contribution is the schedule-initializer benchmark layer, not a replacement for production sequential-convex, collocation, or indirect low-thrust tools. Those mature solvers are downstream consumers and related-work comparators for a schedule or continuous-control initial guess. The controls used here–all-windows continuous refinement, direct shooting, compact collocation, continuation, and tail-coast diagnostics–are included to isolate whether the discrete schedule initializer adds value before handing the problem to stronger continuous machinery.

This framing is a natural fit for an astrodynamics methods journal because it gives the quantum component a real constrained-search role while keeping trajectory-design claims bounded. The artifact is a cislunar trajectory-optimization resource: it exposes schedule-initializer failure modes, continuous-backend sensitivity, missed-thrust accounting, replay provenance, and force-model stress boundaries in a way that downstream astrodynamics solvers can use as controlled initialization tests.

The claims are intentionally narrow. The experiments use normalized CR3BP dynamics, screening thresholds, configured outage families, and research-grade continuous backends. They do not establish quantum advantage, high-fidelity mission validation, fuel-optimal recovery, certified flight design, or universal missed-thrust robustness.

2. Related Work

2.1. Low-thrust trajectory optimization

Low-thrust trajectory optimization has a mature literature spanning indirect methods, direct transcription, collocation, global search, sequential convex programming, and hybrid approaches. Morante et al. survey low-thrust optimization as a hybrid optimal-control problem with both continuous and discrete structure [5]. In the Earth–Moon system, Ozimek and Howell computed low-thrust transfers to libration-point orbits and showed the role of CR3BP structure in preliminary design [6]. Pritchett, Howell, and Grebow used collocation, trajectory stacking, orbit chaining, and continuation to compute low-thrust transfers in the restricted three-body problem, including NRHO-to-DRO examples [7]. More recent cislunar and deep-space solvers include successive convex optimization for libration-point transfers [8], convex-programming approaches for rapid low-thrust trajectory design [9], thrust-regularized convex optimization [10], indirect forward-backward shooting in complex dynamics [11], and adaptive-mesh sequential convex programming [12]. These methods are mature trajectory-design baselines; the present work is not proposed as a replacement for them.

For preliminary cislunar design, the Circular Restricted Three-Body Problem is widely used because it captures the nonlinear Earth–Moon dynamical environment while keeping algorithmic studies tractable. The NASA/JPL Three-Body Periodic Orbits API provides normalized periodic-orbit states, periods, Jacobi constants, stability indices, and system constants for families such as halo orbits and distant retrograde orbits [13]. The catalog-derived benchmark in this paper uses those states as transparent boundary-condition seeds. Its purpose is to expose initialization and recovery failure modes, not to certify an operational transfer.

2.2. Robust and missed-thrust trajectory design

Missed-thrust design has been studied through margin analysis, expected-thrust formulations, virtual recovery trajectories, robust optimal control, and branching trajectory methods. Rubinsztejn et al. embed expected thrust fraction into deterministic trajectory design and report improved resilience to missed-thrust scenarios [14]. Venigalla et al. formulate multi-objective low-thrust trajectory optimization with missed-thrust recovery margin [15]. Grebow and McCarty specifically study low-thrust cislunar transfers, including NRHO-to-DRO transfers, and optimize branching trajectories for outage recovery [1]. Sinha and Beeson focus on initial-guess generation for robust low-thrust missed-thrust design and compare conditional and nonconditional global-search strategies [2]. Robust cislunar low-thrust optimization has also advanced through sequential convex programming and covariance steering; Kumagai and Oguri use chance-constrained covariance steering to design nominal trajectories and correction policies under uncertainty in the Earth–Moon CR3BP [16]. Broader uncertainty-aware low-thrust design includes nonlinear programming and forward-backward shooting under uncertainty [17], and missed-thrust robustness has recently been addressed with sequential convex programming [18]. State-dependent trust-region ideas for autonomous spacecraft SCP further illustrate the level of numerical machinery expected in robust flight-dynamics workflows [19]. These works set a high bar for any initializer: a useful discrete schedule must ultimately feed a continuous optimizer that can produce robust feasible trajectories.

2.3. Quantum schedule initializers

Quantum annealers and QAOA target discrete optimization models such as Ising Hamiltonians and QUBOs. A QUBO minimizes

$$E(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j, \qquad x_i \in \{0,1\}, \tag{1}$$

which is convertible to the Ising form used in quantum annealing and many gate-model workflows [20, 21]. Farhi et al. introduced QAOA as a gate-model variational algorithm for approximate combinatorial optimization [22]. Hadfield et al. generalized this idea to the quantum alternating operator ansatz, where mixers can be designed to preserve hard constraints and evolve only within a feasible subspace [23]. Warm-started QAOA variants further show how a classical prior or relaxation can initialize a more informative quantum state before variational mixing [24]. Grover search, tight quantum-search bounds, amplitude amplification, and quantum minimum finding provide the theoretical search backdrop for the feasible-subspace protocol [25, 26, 27, 28]. Shukla and Vedula formulated trajectory optimization as a discrete search problem and argued that quantum search can provide speedups for suitably discretized cases [29]. De Grossi et al. provide the closest low-thrust QUBO precedent [3]. This paper uses simulated quantum schedule initializers with a narrower hierarchy: the main result is online threshold feasible-subspace amplitude amplification, the earlier top-k Grover rows are oracle diagnostics, and QUBO/QAOA rows are auxiliary diagnostics for the discrete initialization layer. The continuous trajectory remains a classical optimal-control problem.

2.4. Benchmark gap

The targeted catalog benchmark in this paper is deliberately harder than the teacher-controlled case because cislunar transfers between stable or nearly stable periodic orbits are known to be sensitive to the continuous initial guess. The gap addressed here is therefore diagnostic rather than algorithmic replacement: the paper asks whether an online threshold search over constraint-preserving schedule states can reduce explicit schedule evaluations relative to uniform random sampling under a matched total budget, and how the resulting schedule-initializer evidence should be bounded once continuous recovery backends and screening objectives enter the picture. Existing missed-thrust and robust cislunar studies optimize trajectories, recovery branches, correction policies, or mission-design workflows. This paper instead packages reusable source states, masks, configurations, recorded controls, and evidence semantics so those stronger trajectory-design tools can be compared against a transparent initialization benchmark. Production SCP, collocation, and indirect tools remain the appropriate downstream solvers for mission-quality design. The all-windows continuous row and the continuous-backend diagnostics are the relevant controls for the trajectory layer, while uniform random sampling is the primary fair comparator for the online threshold schedule search. The resulting artifact package complements production-oriented studies by making failure modes, claim boundaries, and reproducibility provenance explicit.

3. Problem Formulation

3.1. Forced CR3BP dynamics

The spacecraft state is

$$s = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \tag{2}$$

in the normalized Earth–Moon rotating frame. With mass parameter μ, distances to the primary and secondary bodies are

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \tag{3}$$

$$r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}. \tag{4}$$

The forced CR3BP equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3} + a_x, \tag{5}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3} + a_y, \tag{6}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3} + a_z. \tag{7}$$

The transfer duration T is divided into N uniform segments. A binary variable $x_i \in \{0, 1\}$ gates thrust availability during segment i. During the discrete search stage, a deterministic target-seeking steering law maps the binary schedule to a propagated trajectory,

$$g(s) = K_r(r_f - r) + K_v(v_f - v), \qquad a_i(s) = x_i a_{\max} \frac{g(s)}{\|g(s)\| + \epsilon}. \tag{8}$$

This steering law is not claimed to be optimal. It creates a repeatable map from a bitstring to a trajectory so that discrete schedules can be ranked before continuous refinement. A separate oracle diagnostic, introduced only for the teacher-controlled benchmark, initializes refinement from the hidden continuous controls used to generate the target. That diagnostic is not a competing initializer; it tests whether the refinement formulation can recover the known feasible trajectory when continuous-control initialization is no longer the bottleneck.

3.2. Missed-thrust masks and robust binary cost

Let $\mathcal{M}$ denote a set of outage masks. Each mask zeros thrust in a one- or two-segment contiguous window. For a nominal schedule x, the outage-realized schedule is $x \odot m$. The scaled terminal error under mask m is

$$e_m(x) = \left\| \begin{bmatrix} (r_T(x \odot m) - r_f)/\ell_r \\ (v_T(x \odot m) - v_f)/\ell_v \end{bmatrix} \right\|_2 . \tag{9}$$

The trajectory-evaluated robust binary objective is

$$\begin{aligned} J(x) = {} & w_n e_0(x) + w_w \max_{m \in \mathcal{M}} e_m(x) + w_d \left(\max_{m \in \mathcal{M}} e_m(x) - e_0(x) \right) \\ & + w_p \frac{1}{N} \sum_{i=1}^{N} x_i + w_s \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i| . \end{aligned} \tag{10}$$

The terms penalize nominal miss distance, worst outage miss distance, missed-thrust degradation, active thrust fraction, and thrust/coast switching.

4. Initializer Families

4.1. QUBO surrogate

Because $J(x)$ depends on nonlinear propagation and outage evaluation, it is not itself a QUBO. The method therefore fits

$$\widehat{J}(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j \tag{11}$$

from trajectory-evaluated samples $\{x^{(k)}, J(x^{(k)})\}_{k=1}^{K}$ using ridge-regularized least squares:

$$\min_\theta \sum_{k=1}^{K} \left(J(x^{(k)}) - \widehat{J}(x^{(k)}; \theta) \right)^2 + \lambda \|\theta\|_2^2 . \tag{12}$$

The fitted QUBO is then sampled by classical QUBO simulated annealing or converted to a diagonal cost Hamiltonian for statevector QAOA. The present experiments use QAOA only as a small simulated gate-model sampler; no hardware quantum advantage is implied.

4.2. Discrete initializers

The experiments compare random search, cross-entropy search, a genetic algorithm, true-objective simulated annealing, surrogate QUBO simulated annealing, and statevector QAOA. Random search and true-objective simulated annealing evaluate propagated schedules directly. Cross-entropy updates Bernoulli segment probabilities from elite samples. The genetic algorithm uses elitism, crossover, and mutation. The QUBO and QAOA methods use true trajectory evaluations to train $\widehat{J}$, then validate their best sampled schedules with the true objective.

Budget accounting is therefore reported in two ways: solver true evaluations exclude shared QUBO training, while total true evaluations include QUBO training. Equal total-evaluation comparisons are the relevant fair comparison.

4.3. Feasible-subspace amplitude amplification

The structured quantum-search experiment isolates recorded cardinality subspaces from the phase-shift cardinality benchmark. For the primary one-coast case, let $\mathcal{F} = \{x \in \{0,1\}^{12} : \sum_i x_i = 11\}$ be the feasible schedule set. The 12 basis states are ordered by the single coast-window index. The generalized runner can instead target any named cardinality group in `data/results/phase_shift_cardinality_ablation/cardinality_refinements.csv`; this paper also reports the 66-state `k10_full` group with two coast windows. In each case, the cost assigned to a basis state is the recorded branch-refined selected-worst error.

For the online threshold experiment, the finite feasible subspace is indexed as $\{|j\rangle : j = 0, \ldots, M-1\}$, where the classical lookup table maps j to one admissible binary schedule in $\mathcal{F}$. The implemented threshold oracle is the finite-table comparator

$$O_\tau |j, b\rangle = |j, b \oplus [C_j < \tau]\rangle, \tag{13}$$

with τ set to the current incumbent cost. This is a query-model benchmark, not a circuit or hardware proposal. A synthesized implementation would still need finite-subspace state preparation, qROM/qRAM or table lookup for C_j, a reversible fixed-point comparator for $C_j < \tau$, and, if costs were not precomputed, a reversible cost oracle for trajectory propagation and branch recovery. This paper performs none of those resource estimates and reports only threshold-query counts and explicit candidate lookup/evaluation counts.

The experiment assumes exact finite-subspace state preparation and reflection about the prepared state, equivalently a qROM/table-oracle model for the precomputed C_j values. It does not implement a reversible trajectory propagator, a reversible least-squares branch-refinement oracle, or any hardware-level data-loading circuit. Ideal BBHT/Durr–Hoyer style bounds therefore apply only to threshold-oracle calls under that implemented finite-table oracle: if r states are below the incumbent threshold, a successful threshold search step has the familiar $O(\sqrt{M/r})$ query scale, and minimum finding has $O(\sqrt{M})$ threshold-query scaling up to constants in the ideal oracle model. The wall-clock cost of this paper is instead

$$T_{\text{total}} = T_{\text{table}}(M) + Q_\tau T_{O_\tau} + N_{\text{eval}} T_{\text{lookup}} + N_{\text{refine}} T_{\text{branch}}, \tag{14}$$

where T_{table} is classical table construction by recorded branch-refined costs or feedback propagation, Q_τ is the counted threshold-oracle usage, N_{eval} is the counted explicit candidate-cost lookup/evaluation count, and $N_{\text{refine}} T_{\text{branch}}$ is the downstream classical branch-recovery cost for selected schedules. Thus the query accounting separates the quantum front-end oracle model from classical trajectory propagation and refinement; no wall-clock, hardware, or reversible-trajectory-oracle advantage is claimed.

Costs are normalized as

$$\tilde{C}_j = \frac{C_j - \min_k C_k}{\max_k C_k - \min_k C_k}. \tag{15}$$

The warm-start probabilities are

$$p_j^{(0)} = \frac{(\tilde{C}_j + \epsilon)^{-1}}{\sum_k (\tilde{C}_k + \epsilon)^{-1}}, \qquad \epsilon = 0.02, \tag{16}$$

and the initial state is $|\psi_0\rangle = \sum_j \sqrt{p_j^{(0)}} |j\rangle$. A threshold oracle marks

$$\mathcal{G}_\tau = \{ j \in \mathcal{F} : C_j \le \tau \}, \tag{17}$$

and the top-k oracle marks the k feasible schedules with smallest recorded C_j values. For a chosen good set $\mathcal{G}$, one exact Grover-style iterate is

$$Q_{\mathcal{G}} = (2|\psi_0\rangle\langle\psi_0| - I) O_{\mathcal{G}}, \qquad O_{\mathcal{G}} |j\rangle = \begin{cases} -|j\rangle, & j \in \mathcal{G}, \\ |j\rangle, & j \notin \mathcal{G}. \end{cases} \tag{18}$$

The reported state after r iterations is $|\psi_r\rangle = Q_{\mathcal{G}}^r|\psi_0\rangle$. In the revised headline protocol, however, $\mathcal{G}$ is not a recorded top-k set. The online threshold search samples and evaluates an initial incumbent $j^\star$, sets the threshold $\tau = C_{j^\star}$, marks only states satisfying $C_j < \tau$ through O_τ, applies a seeded BBHT-style random number of Grover iterations, samples one candidate, and evaluates that candidate classically by table lookup. The incumbent changes only if the sampled candidate has lower evaluated cost; otherwise the Grover-window parameter grows up to an $O(\sqrt{M})$ cap. The simulator records incumbent and candidate cost evaluations, threshold-oracle calls, total oracle calls, best-found index/cost, global-optimum success, and top-percentile hit flags. Classical uniform-without-replacement is the primary matched-budget baseline. The ranked screening row is reported under the same schedule-evaluation count only as a ceiling/diagnostic ranking: in the feedback-screening cases it is driven by a closely related feedback worst-error score and should be read as near-perfect ranking information, not as a fair deployable classical baseline.

The earlier top-2 protocol is retained as a diagnostic ablation, not the main algorithm. It marks the two feasible schedules with smallest recorded selected-worst errors, or all states if the feasible subspace has fewer than two states. Let $a_0 = \sum_{j\in\mathcal{G}} p_j^{(0)}$ be the initial warm-start probability on this diagnostic good set. Following the standard amplitude-amplification success expression [25, 26, 27], the iteration count is chosen as

$$r^\star = \arg\max_{r\in\{0,\ldots,12\}} \sin^2\!\left((2r+1)\arcsin\sqrt{a_0}\right), \tag{19}$$

with ties broken toward the smaller r. This rule uses only the predeclared top-2 oracle, the warm-start distribution, and the fixed iteration cap; it does not choose an iteration count after comparing benchmark outcomes.

The simulations use exact dense linear algebra over $\mathcal{F}$, with no sampling noise, no hardware noise model, and no infeasible bitstrings. A separate exploratory runner sweeps threshold quantiles $\{0, 0.10, 0.25, 0.50\}$, top-k values $\{1, 2, 3, 5, 8\}$ when available, and $r = 0, \ldots, 12$. Each warm-started sweep row is paired with a matched uniform feasible-subspace comparator using the same good set and iteration count. The top-2 oracle uses recorded branch-refined costs and is therefore a diagnostic only. The online threshold protocol is the implemented minimum-finding-style schedule search [28]; in simulation the threshold oracle is evaluated from the finite cost vector, while query accounting separates threshold-oracle calls from explicit schedule-cost evaluations.

The earlier structured-QAOA package is retained as an auxiliary comparison. It uses the same feasible cardinality bases and recorded selected-worst costs, but evolves with an adjacency-based feasible-subspace mixer and optimized QAOA angles rather than an explicit Grover oracle/reflection iterate. Those rows are useful supporting evidence, not the headline quantum algorithm in this revision.

4.4. Continuous branch-recovery refinement

The refinement stage converts a selected binary schedule into a continuous low-thrust candidate by optimizing piecewise-constant acceleration vectors only in active windows. Refined accelerations are projected to the Euclidean thrust ball $\|u_i\|_2 \leq a_{\max}$ before propagation and persistence. The branch-recovery mode optimizes a nominal control sequence plus selected-outage-specific post-outage recovery controls. Its least-squares residual includes nominal terminal error, selected outage terminal error, fuel regularization, control regularization, and inter-segment smoothness. A run is counted as successful only if both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds.

The paper also reports an all-mask diagnostic: after refinement on selected outages, every one- and two-segment outage mask is evaluated without claiming that all masks were optimized. This separates selected-branch recovery success from universal outage robustness.

5. Experimental Setup

5.1. Common dynamics and source state

All experiments use normalized Earth–Moon CR3BP dynamics with $\mu = 0.01215058560962404$. The common initial state is taken from the first row of an Earth–Moon L2 southern halo query with period in $[1.55, 1.65]$ TU from the NASA/JPL periodic-orbit API:

$$s_0 = [1.0324985738, 0, -0.1884844917, 0, -0.1249781287, 0]. \tag{20}$$

This state is NRHO-like in the sense that it lies in the L2 southern halo family, but the paper treats it as a normalized benchmark state, not a certified operational NRHO.

5.2. Dimensional scale and threshold interpretation

The normalized Earth–Moon scale is used only to interpret orders of magnitude. Taking the public NASA average Earth–Moon distance as 1 LU $\approx$ 384,400 km [30], the recorded CR3BP normalization gives 1 TU $\approx$ 375,190.259 s = 4.34248 days, velocity unit $\approx$ 1.02455 km/s, and acceleration unit $\approx$ 2.73074 mm/s^2. Thus the main phase-shift transfer time $T = 0.5$ TU is about 2.17 days, while the hard-catalog tail-coast recovery case with $T = 4.0$ TU is about 17.37 days. The acceleration bounds used in the phase-shift and high-thrust diagnostics correspond to $a_{\max} = 0.2 \approx 0.546$ mm/s^2 and $a_{\max} = 1.0 \approx 2.731$ mm/s^2, respectively.

The terminal-error thresholds are internal benchmark screening tolerances in the normalized metric

$$\sqrt{\|\Delta r\|_2^2 + \|\Delta v\|_2^2/0.35^2}, \tag{21}$$

not flight navigation or operational targeting tolerances. If velocity error is zero, the nominal threshold 0.09 gives a component upper-bound position-only error of about 34,596 km, and the selected-recovery threshold 0.17 gives about 65,348 km. If position error is zero, the same thresholds give component upper-bound velocity-only errors of about 32.3 m/s and 61.0 m/s under the configured velocity scale. Mixed position/velocity errors share the same normalized budget, so these dimensional values should not be read as simultaneous tolerances. This scale is the reason the manuscript frames the results as preliminary design and initializer benchmarking rather than operational targeting.

Practically, the loose thresholds are stage-gate and ranking tolerances for initializer benchmarking. A mission-design workflow would take any schedule that passes these gates into tighter targeting, higher-fidelity propagation, and continuous optimal-control refinement before it could support an operational claim. The practical claim here is therefore reduced candidate-discovery burden, improved ranking, or higher downstream optimizer convergence probability under a fixed evaluation budget, not operational terminal accuracy. The threshold-sensitivity and replay/stress artifacts are included to expose how quickly this initializer evidence changes when the gate or model is tightened.

5.3. Benchmark families

Two benchmark families define the main evidence path. The harder catalog-derived family uses a distant retrograde orbit seed from an Earth–

Moon DRO query,

$$s_{\mathrm{DRO}} = [0.8161260827, 0, 0, 0, 0.5076556931, 0], \tag{22}$$

and generates the target by ballistic propagation of that seed. The original catalog-derived schedule run uses $T = 2.7$ TU, $N = 12$, $a_{\max} = 0.055$, and one- and two-segment outage masks. It remains negative evidence in the supplement. The main hard-catalog result instead reports a later locked-nominal tail-coast continuous-backend diagnostic at $T = 4.0$ TU, $N = 14$, and $a_{\max} = 1.0$, where the final five nominal control segments are constrained to zero and all configured one- and two-segment masks are selected and evaluated in one fixed-final-time row.

The non-teacher phase-shift family is derived from the same JPL L2 southern halo source state s_0. The target is generated by zero-thrust CR3BP propagation of s_0 for a phase time of 0.3 TU, while the transfer must arrive at that target after $T = 0.5$ TU. A zero-thrust trajectory has normalized terminal error 0.3758, so the case is not a do-nothing pass. The primary initializer benchmark uses $N = 12$, $a_{\max} = 0.2$, one-segment outage masks, and a duty-cycle prior

$$J_\rho(x) = w_\rho \left(\frac{1}{N} \sum_{i=1}^{N} x_i - \rho \right)^2, \tag{23}$$

with $\rho = 11/12$ and $w_\rho = 12$. This prior is not an oracle trajectory correction; it encodes a preference for one coast window while preserving a dense low-thrust arc.

The supplement preserves the secondary diagnostic families: the catalog pilot and QUBO fit diagnostics, feasibility and direct multiple-shooting stress tests, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, dense-repair and legacy cardinality ablations, and the teacher-controlled feasible benchmark. These diagnostics explain provenance and failure modes but are not promoted to headline evidence in the main article.

5.4. Metrics

The primary metrics are refined nominal terminal error, refined selected missed-thrust worst error, all-mask diagnostic worst error, schedule active fraction, nominal and recovery fuel, total true trajectory evaluations, and refinement success rate. A run is counted as a robust refinement success only when both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds. Unless otherwise stated, those thresholds are 0.09 for nominal success and 0.17 for selected robust recovery success.

The main $N = 12$ duty-cycle-prior statistics package repeats seven methods over 30 deterministic seeds, for 210 raw method–seed rows, and reports Wilson success intervals, paired per-seed success differences, bootstrap confidence intervals for mean selected-worst-error differences, and exact two-sided sign tests. A separate focused QAOA-depth package repeats the QUBO-derived methods over 30 deterministic seeds, for 150 raw method–seed rows. The hard-catalog tail-coast row `tail_coast_all_one_two_segment_t5_portfolio` uses the same thresholds, selects all configured one- and two-segment masks in one case row, and reports optimizer convergence separately from threshold feasibility so late-tail direct evaluations without recovery variables are not mistaken for optimizer-converged branch solves.

6. Results

The main results are screening evidence for a normalized CR3BP benchmark. The dimensional caveat in Section 5.2 applies to every table: the 0.09/0.17 thresholds correspond to large position-only or velocity-only errors and should not be read as operational targeting tolerances. Secondary tables and figures are retained in the supplement so that the main article follows the primary claim path, while the evidence-synthesis table below cross-indexes representative rows from those recorded artifacts.

6.1. Claim hierarchy

Table 1 records the claim boundary used throughout the paper. It lists each headline claim, its primary artifacts, its status, and interpretations that are explicitly prohibited by the evidence.

6.2. Claim-ledger summary

The generated claim ledger in `data/results/claim_evidence_ledger/` is retained as a reviewer-facing audit artifact rather than as a dense main-text table. It reads recorded CSV/JSON files only and separates selected-branch evidence, all-mask diagnostics, all-configured-mask rows, accepted-control replay, force-model contrast, stress probes, and retuned-recovery evidence. The full ledger, tail-coast threshold audit, and tail-coast branch audit are reported in the supplement. The main text keeps the compact hierarchy in Table 1 and the cross-backend synthesis in Table 2; together they state the operative boundaries without turning the article into an artifact listing.

For practitioners, the reporting lesson is concrete: include the all-windows continuous control, do not collapse selected-branch and all-mask semantics, preserve negative QAOA/QUBO evidence, and archive provenance in machine-readable artifacts. In the current ledger, the new independent-midpoint Hermite–Simpson headroom package is all-configured one-segment normalized-CR3BP evidence with separate branch-control replay, simple bicircular phase-stress, cached-Horizons-derived solar-tidal replay, single- plus multi-epoch cached-Horizons point-mass retuning rows, and a SPICE-derived replay row, while the earlier independent-HS p=0.4 baseline row remains selected-branch/all-mask diagnostic evidence.

6.3. Cross-backend evidence synthesis

Table 2 is generated by `scripts/run_evidence_synthesis.py`. The script reads only recorded CSV/JSON artifacts and writes `data/results/evidence_synthesis/evidence_synthesis.csv`, `data/results/evidence_synthesis/evidence_synthesis_metadata.json`, and the LaTeX table used here. It is not a new optimization run. Its purpose is to put the broader validation axes into the main claim path without overstating them.

The synthesis clarifies the positive contribution. The 30-seed threshold-sensitivity row shows that at the tight $(0.050, 0.090)$ check every sampled method is 0/30 while all-windows continuous remains 30/30, so the paper does not claim sampler superiority. The continuation-extension rows show that all-mask direct multiple-shooting continuation can meet a tighter $(0.065, 0.100)$ phase-shift screen for all one-segment masks and for all configured one/two masks, but not the 0.05 nominal headroom check. The direct-collocation and earlier independent-midpoint Hermite–Simpson rows show compact continuous backends with tight selected/all diagnostics while preserving selected-branch semantics. The new independent-midpoint Hermite–Simpson headroom package upgrades that phase-shift evidence to all-configured one-segment rows at phase time 0.4 and $a_{\max} = 0.2$: all eight configured masks are selected and evaluated, with the original row at nominal error 0.0111 and selected/all worst error 0.0774, and a polished converged row at nominal error 0.0113 and selected/all worst error 0.0779. A separate simple bicircular phase-sweep stress replay validates the persisted polish-row controls over eight Sun phases: all 8 nominal phases and all 64 branch-phase checks pass, with maximum nominal error 0.022138676654057693 and maximum branch-phase error 0.08557051343145317. A cached-Horizons-derived solar-tidal replay for the same polish row uses a representative 2026-Jan-01 JPL Horizons geometry cache and also passes, with nominal error 0.018363195236986728, branch worst 0.07422350563850917, and branch pass count 8/8. A stronger cached-Horizons Earth/Moon/Sun point-mass model at the same representative epoch records a negative direct replay for persisted controls, with nominal error 0.3812580376880591 and branch worst 0.3797450961017463, but independent least-squares retuning restores feasibility with retuned nominal error 0.02143944130524006, retuned branch worst 0.02473065115224942, and branch pass count 8/8. The four-epoch multi-epoch retuning wrapper keeps the retuned branch pass count at 32/32, and the SPICE-derived replay of those 36 retuned controls keeps nominal/branch worst errors at 0.021439441253166033/0.024730650824609506 with maximum delta $3.2763991519857427 \times 10^{-10}$ relative to the Horizons-retuned replay. The hard-catalog rows show that the fixed-final-time pass is scoped to the tail-coast locked-nominal backend; an independent-HS selected-outage hard-catalog diagnostic remains a negative result.

6.4. Tail-coast hard-catalog diagnostic

Table 3 reports the strongest hard-catalog fixed-final-time result. The setup solves and freezes a locked nominal control, then refines it with the final five nominal control segments constrained to exact zero. Branches keep the original target and original final time; no delayed target or added recovery horizon is used. The nominal-only provenance row has tail-coast nominal error 0.022992 and therefore passes the nominal screening threshold, while its masked-nominal all-mask diagnostic remains 27.8351 because no branch recovery is run.

The headline row is `tail_coast_all_one_two_segment_t5_portfolio`. It sets `outage_lengths=[1,2]`, selects all configured masks, and evaluates all 27 selected masks in one fixed-final-time case row. The row reaches tail-coast nominal error 0.02299233817855882, selected fixed-time worst error 0.0936063931709301, and all-mask fixed-time diagnostic worst error 0.0936063931709301, so it meets the configured 0.09/0.17 screening thresholds. The branch portfolio evaluates deterministic fallback starts for four branches, charges 5929 function evaluations, and records runtime about 1833.6 s in the current CSV. Among branches with post-outage recovery variables, the optimizer-converged count is 25/25. The remaining two late-tail masks have no post-outage recovery variables; they are direct fixed-final-time threshold evaluations rather than optimizer-converged branch solves. Therefore `branch_optimizer_all_success=false` remains a conservative CSV accounting flag even though `meets_thresholds=true`.

This result is deliberately scoped. It is fixed-final-time continuous-backend evidence with a locked nominal control, final-five-segment coast, configured one- and two-segment masks, fallback starts, and late-tail no-recovery-variable accounting. It is not a binary-sampler result, not QUBO/QAOA evidence, not fuel optimality, not high-fidelity validation, and not robustness to outage families beyond the configured masks.

6.5. Replay and external-validity boundary

The full accepted-control replay, nominal sidecar replay/stress, independent-HS branch-control replay, independent-HS cached-Horizons solar-tidal replay, independent-HS cached-Horizons point-mass retuning, independent-HS SPICE-derived ephemeris replay, Horizons contrast, bicircular stress, and bicircular retuning tables are moved to the supplement. The main facts are concise. The independent-HS branch-control replay contains two p=0.4 nominal rows and 16 branch rows; repropagation of persisted endpoint and midpoint schedules under the configured normalized CR3BP model gives zero nominal, branch, selected-worst, and all-mask-worst terminal-error deltas at tolerance 10^{-10}. The focused accepted-control replay for `tail_coast_all_one_two_segment_t5_portfolio` contains one nominal row and 27 branch rows; repropagation under the configured normalized CR3BP model gives zero nominal, branch, and maximum terminal-error delta at tolerance 10^{-10}. Both replay packages are deterministic replay of persisted controls, not new solves or mature-backend portability/bridge checks.

The external-validity checks are deliberately mixed and conservative. A cached JPL Horizons contrast for the same hard-catalog window samples 15 transfer nodes from a fixed 2026-Jan-01 TDB epoch and records Earth–Moon distance/rate variation, Sun distance/phase differences, and solar-tidal acceleration differences; it is a force-model contrast only, not SPICE validation or ephemeris propagation. The converged independent-HS all-configured polish row provides a positive deterministic stress-probe result under the simple circular solar-tidal model: the persisted endpoint-plus-midpoint controls pass all 8 nominal Sun phases and all 64 branch-phase checks with maximum errors 0.022138676654057693 and 0.08557051343145317. The same polish row also passes a cached-Horizons-derived solar-tidal replay at the representative 2026-Jan-01 epoch, with nominal error 0.018363195236986728, branch worst 0.07422350563850917, branch pass count 8/8, CR3BP replay delta 0.0, and Sun distance range 382.6857920288508–382.693044178952 LU. This uses cached JPL Horizons geometry but remains a simplified stress probe, not SPICE/high-fidelity/flight validation. Under a stronger geocentric Earth/Moon/Sun point-mass model using the same cache, direct replay of the persisted polish controls fails with nominal error 0.3812580376880591 and branch worst error 0.3797450961017463. Independent least-squares retuning of the nominal and branch endpoint-plus-midpoint controls restores feasibility at that representative epoch, with retuned nominal error 0.02143944130524006, retuned branch worst 0.02473065115224942, branch pass count 8/8, optimizer success count 9/9, and cache SHA-256 prefix `13fe6993`; the full hash is retained in the supplement and metadata. The four-epoch cached-Horizons wrapper repeats that point-mass retuning protocol at Jan/Apr/Jul/Oct 2026, and the SPICE-derived replay then repropagates the 36 already-retuned controls with compact Moon/Sun vector caches derived from `naif0012.tls`, `de442s.bsp`, and `gm_de440.tpc`: all 4 nominal rows and all 32 branch rows pass, worst nominal/branch errors are 0.021439441253166033/0.024730650824609506, and the maximum delta from the Horizons-retuned replay is $3.2763991519857427 \times 10^{-10}$. This is a point-mass ephemeris-source replay package, not full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, or quantum evidence. In contrast, replaying the hard-catalog accepted controls with a simple bicircular solar-tidal term fails the configured thresholds: every nominal solar-tidal phase fails the 0.09 nominal threshold, only 22/108 branch-phase rows pass the 0.17 branch threshold, and the maximum solar-tidal terminal error is 37.7596. A fixed-phase retuning batch under the same simple bicircular model also remains negative: retuned nominal error is 0.316772, configured branch pass count is 19/27, maximum retuned branch error is 6.0299, and strict branch pass count is 16/27. The hard-catalog tail-coast pass is therefore foregrounded as normalized-CR3BP evidence only, while the independent-HS stress, retuning, and ephemeris-source replay results are representative-epoch probes rather than high-fidelity validation.

6.6. Online threshold feasible-subspace search

The headline quantum-search experiment is regenerated with `scripts/run_online_quantum_search.py`. It uses two evidence tiers. The `recorded_branch_refined` tier reads the existing branch-refined selected-worst costs for `one_coast_k11_full` ($M = 12$) and `k10_full` ($M = 66$). The `feedback_screening` tier enumerates larger high-duty CR3BP feedback-propagation subspaces without continuous refinement: $N = 16, k = 12$ ($M = 1820$) and $N = 20, k = 16$ ($M = 4845$). These larger rows use the configured phase-shift feedback objective and terminal errors only; they are not branch-refined trajectory evidence, high-fidelity evidence, or flight validation.

Table 5 reports 200 deterministic seeds at total-oracle budgets $\lceil\sqrt{M}\rceil$, $2\lceil\sqrt{M}\rceil$, $4\lceil\sqrt{M}\rceil$, and M. Budget labels are retained after capping numeric budgets at M, so very small subspaces can contain duplicate numeric budgets, for example one-coast $4\lceil\sqrt{M}\rceil$ and M both equal M, and those rows should not be read as independent regimes. The deployable classical comparators are matched-budget direct finite-subspace searches over the same precomputed constrained cost vectors: uniform random sampling without replacement, simulated annealing with schedule Hamming-swap neighbors, cross-entropy with repaired Bernoulli schedule models, and genetic search with schedule crossover/mutation/repair. They do not regenerate trajectories beyond the existing subspace cost construction. Summary rows include Wilson 95% success intervals, and Table 6 reports paired online-minus-baseline comparisons by case, budget, and seed. Negative best-cost differences favor online threshold search.

For `k10_full`, online threshold amplitude amplification improves global-optimum hit rate over all four deployable baselines at the three sublinear budgets: 0.235/0.62/0.935 for online threshold search versus 0.07/0.20/0.58 for uniform random, 0.10/0.22/0.465 for simulated annealing, 0.16/0.32/0.595 for cross-entropy, and 0.16/0.355/0.625 for the genetic baseline. The paired mean best-cost differences at $4\lceil\sqrt{M}\rceil$ are small but favor online against the deployable baselines, with bootstrap 95% intervals below zero for simulated annealing, cross-entropy, and genetic search. At the full $M = 66$ budget, uniform random necessarily enumerates the finite subspace and reaches the optimum in all seeds, while online misses once and should not be read as superior to enumeration.

The feedback-screening rows are larger but weaker trajectory evidence. On $M = 1820$, online threshold search improves hit rates over uniform, simulated annealing, cross-entropy, and genetic baselines at $4\lceil\sqrt{M}\rceil$: 0.86 versus 0.075/0.50/0.52/0.815, with a paired mean best-cost difference of -5.0×10^{-4} versus genetic search and a bootstrap interval that crosses zero. On $M = 4845$, online is much stronger than uniform, simulated annealing, and cross-entropy at $4\lceil\sqrt{M}\rceil$ (0.875 versus 0.06/0.33/0.60), but the genetic baseline reaches 0.935 and a slightly lower mean best cost; the paired cost difference is $+3.76\times10^{-4}$ with bootstrap interval $[+1.5\times10^{-5}, +7.36\times10^{-4}]$. Thus the online protocol is a competitive threshold-only constrained-search mechanism, not a universal replacement for tuned classical finite-subspace heuristics. The $M = 12$ one-coast family is too small to support a strong sublinear story: online threshold search is close to uniform random at low budgets and exhaustive uniform sampling necessarily wins at M.

The ranked baseline in Table 5 is a screening diagnostic, not a quantum comparator. For feedback-screening rows it uses a closely related feedback worst-error score from the same CR3BP feedback-propagation screening setup, so it is intentionally optimistic and reaches the best feedback-screening state immediately. It is included to show what a near-perfect classical ranking signal would do under the same evaluation count; it is not a fair deployable baseline that the online protocol is claimed to beat. The practical implication is therefore not that quantum search beats every classical screen. Rather, the online protocol removes the most serious top-2-oracle objection and shows that a threshold-only feasible-subspace search can reduce the number of explicit candidate schedule evaluations needed to find low-cost schedules in constrained families relative to uniform random sampling. The all-windows continuous backend remains the relevant lower-error continuous-control diagnostic outside these constrained schedule families.

To keep the large feedback-screening rows from being overread, a targeted branch-refinement diagnostic reruns the configured branch-recovery backend on only the top feedback-screening schedule from each large case. Table 7 shows the result. The top $N = 16, k = 12$ schedule `0000111111111111` and top $N = 20, k = 16$ schedule `00001111111111111111` both improve substantially from feedback-screening terminal errors after branch recovery, and their selected-worst/all-mask diagnostic errors fall below the configured robust threshold 0.17. Neither row passes the configured nominal threshold 0.09 under the default recovery cap: the branch nominal errors are 0.134195 and 0.115405, with selected-worst errors 0.153803 and 0.131575 and all-mask diagnostics 0.157742 and 0.134621. The $N = 20$ row records optimizer success, while the $N = 16$ row reaches the 35-evaluation cap without optimizer success. This is useful negative evidence: online quantum search is promising as a discrete feedback-screening front end, but the large $M = 1820$ and $M = 4845$ rows remain screening-scale evidence until full branch-refined feasible seeds, or a stronger continuous backend, close the nominal branch threshold.

6.7. Diagnostic predeclared one-coast feasible-subspace search

The dedicated Grover-style experiment reads the recorded one-coast rows from `data/results/phase_shift_cardinality_ablation/cardinality_refinements.csv`. The best one-coast schedule by refined selected-worst error is `111111111110`, with selected-worst error 0.0832614239111052. Uniform sampling over the 12 feasible one-coast states has expected selected-worst error 0.0874612047385159 and assigns probability 1/12 to the best schedule. The inverse-cost warm start alone improves the expected selected-worst error to 0.0842841487476258 and assigns probability 0.47742489298267526 to the best schedule.

The predeclared `dual_candidate_warm_amplification` protocol marks the two one-coast schedules with smallest recorded selected-worst errors. Their initial warm-start mass is $a_0 = 0.6388030841472769$, so the fixed amplitude-amplification rule selects $r = 2$. The resulting exact state assigns probability 0.7423423742484173 to the best schedule, raises marked-set probability to 0.9932674335443776, and reaches expected selected-worst error 0.0833854880397595. Against the matched uniform-Grover top-2, two-iteration comparator, the predeclared row lowers expected selected-worst error by 0.0013185708519839051 and doubles best-schedule probability from 0.3713991769547326 to 0.7423423742484173.

The exploratory sweep contains an equivalent two-mark threshold row, `grover_warm_thresholdle0p0836849565092_r2`, with the same probabilities and expected error. The audit therefore records that the predeclared protocol equals the best warm-started sweep outcome for this family, while still separating the protocol row from the sweep rows. A uniform top-1 oracle row can concentrate even more probability on the best state in this small 12-state space; that is an oracle-sweep diagnostic rather than a warm-start dominance claim.

The all-windows continuous row remains outside the one-coast feasible subspace and still records the lower selected-worst error 0.06006019530605116.

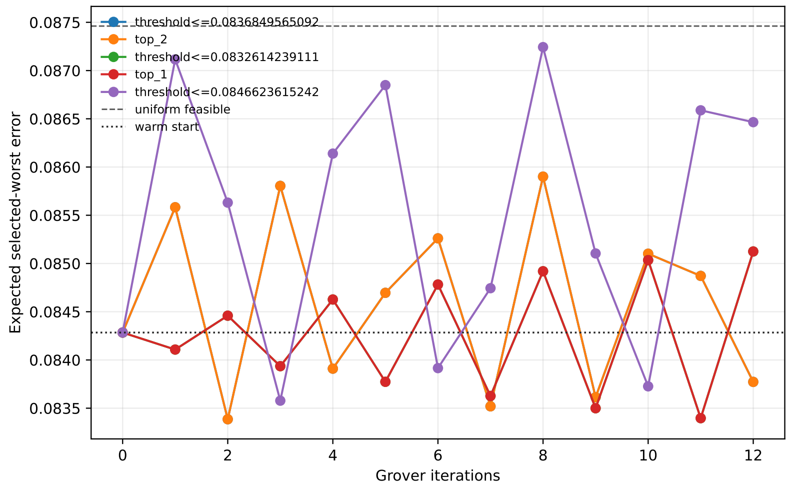

Figure 1: Warm-started one-coast Grover sweep used as an exploratory diagnostic. The predeclared protocol uses the top-2 oracle and the fixed iteration rule rather than choosing this curve after the sweep.

6.8. Diagnostic predeclared `k10_full` feasible-subspace search

The same runner targets the 66 feasible schedules in the `k10_full` group of `data/results/phase_shift_cardinality_ablation/cardinality_refinements.csv`. This is a larger cardinality subspace than the one-coast case: each basis state has 10 active thrust windows and two coast windows. The best recorded `k10_full` schedule is `111111111100`, with coast windows 10 and 11 and refined selected-worst error 0.1055748925448516.

Uniform sampling over the 66 feasible states has expected selected-worst error 0.11375786488534918 and probability $1/66 = 0.01515$ on the best schedule. The inverse warm start alone improves the expectation to 0.108671256502909 and gives best-schedule probability 0.12012421123686709. The predeclared protocol again marks the top two recorded-cost schedules. Their initial warm-start mass is $a_0 = 0.23275348972275725$, so the fixed amplitude-amplification rule selects $r = 1$. The exact top-2 amplified state reaches expected selected-worst error 0.10560395128688618, marked-set probability 0.996348617182711, and best-schedule probability 0.5142161000403436. Relative to the matched uniform-Grover top-2, one-iteration comparator, the predeclared row lowers expected selected-worst error by 0.006293776482407976 and raises best-schedule probability from 0.12556696440993945 to 0.5142161000403436.

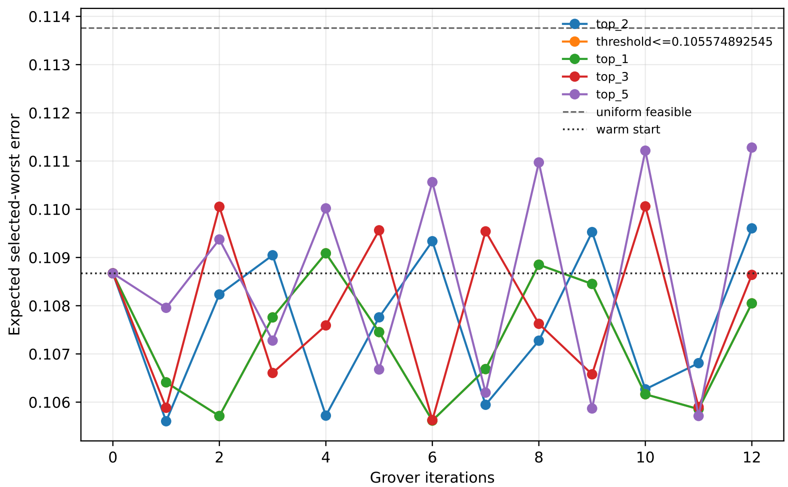

Figure 2: Warm-started `k10_full` Grover sweep used as an exploratory diagnostic. The predeclared protocol uses the same top-2, one-iteration row for this family.

Here the predeclared row is also the best warm-started row in the exploratory sweep.

6.9. Auxiliary structured-QAOA evidence

The structured-QAOA packages are retained as supporting evidence because they use the same feasible cardinality bases and recorded selected-worst costs but a different quantum ansatz. On the one-coast family, the best auxiliary row `warm_start_xy_qaoa_p4` reaches expected selected-worst error 0.08353137637791036 and assigns probability 0.653075051576728 to `111111111110`. The follow-on one-coast sensitivity package in `data/results/structured_qaoa_sensitivity/` shows the same complete/inverse $p = 4$ pattern over path/cycle/complete mixers, inverse/exponential warm starts, depths $p = 1, \ldots, 4$, and restart counts 1/4/8, with small mostly negative structured-minus-matched uniform deltas.

On the larger `k10_full` feasible subspace, the strongest auxiliary structured-QAOA row is `structured_inverse_complete_p4_r8`. It reaches expected selected-worst error 0.10622367343274204, assigns probability 0.179971463256841 to `111111111100`, and beats the matched complete/uniform QAOA comparator by 4.11×10^{-4} in expected selected-worst error. The predeclared Grover-style top-2 row is better on both expected selected-worst error and best-state probability, but it is now only a diagnostic because it uses a recorded-cost top-2 oracle. The comparison to the generic `qaoa_statevector` baseline remains descriptive and not like-for-like, because the generic row is still an unconstrained bitstring summary from the fitted QUBO surrogate.

6.10. Thirty-seed main-method statistics

Tables 16 and 17, together with Figure 3, repeat the configured $N = 12$ duty-cycle-prior method comparison over 30 deterministic seeds. The raw package contains 210 rows, i.e., 30 seeds times seven methods: random search, cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, the configured statevector QAOA sampler, and all-windows continuous refinement.

The success-rate evidence strengthens the 10-seed result but does not create a quantum-performance claim. Random search succeeds in 11/30 seeds, with Wilson 95% interval $[0.219, 0.545]$. Cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement each succeed in 30/30 seeds, with Wilson interval $[0.886, 1.000]$. The configured `qaoa_statevector` method succeeds in 13/30 seeds, with interval $[0.274, 0.608]$.

The paired error comparisons are more important for the aerospace interpretation. Against all-windows continuous refinement, every sampled method has a positive mean selected-worst-error difference; negative would favor the sampled method. The exact two-sided sign-test value is approximately 1.86×10^{-9} for each sampler versus all-windows continuous refinement. Thus, the all-windows continuous baseline is not just equally successful for the strongest methods; it also gives lower selected-worst error on this benchmark. Against surrogate-QUBO simulated annealing, cross-entropy, genetic search, and true-objective simulated annealing tie the success rate and have small nonsignificant selected-worst-error deltas. Random search is worse. The configured statevector QAOA row has lower success and a positive mean selected-worst-error delta of 0.0153 versus surrogate-QUBO simulated annealing, with exact sign-test $p = 0.1849$.

This package therefore strengthens the statistics beyond a QAOA-only ablation while preserving the conservative conclusion. Duty-cycle priors help several samplers find feasible high-duty schedules, but all-windows continuous fuel regularization remains a stronger mechanism for selected-worst error in this benchmark, and the evidence does not support quantum advantage or QAOA superiority.

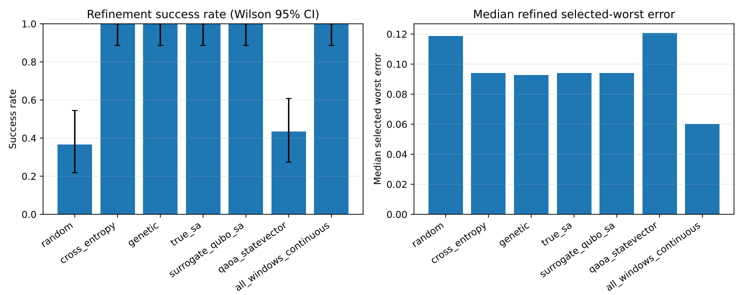

Figure 3: Thirty-seed main-method statistics for the duty-cycle-prior benchmark. High-duty samplers can be feasible, but all-windows continuous refinement remains lower-error on paired selected-worst comparisons.

Table 18 adds a threshold-sensitivity sanity check computed only from `data/results/phase_shift_cardinality_30seed/raw_results.csv`. At the stricter $(0.075, 0.120)$ nominal/selected-worst thresholds, cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement still pass 30/30 seeds, while random search falls to 4/30 and `qaoa_statevector` remains 13/30. At $(0.065, 0.100)$ the sampled high-duty schedules largely stop passing, but all-windows continuous remains 30/30. At the tight $(0.050, 0.090)$ check, every sampled method is 0/30 while all-windows continuous is still 30/30. Thus stricter thresholds do not rescue sampled-method superiority; they quantify the margin limitation of the sampled schedules and reinforce the all-windows continuous lower-error diagnostic.

6.11. Focused QAOA/QUBO ablation

Table 19 and Figure 4 isolate the QUBO-sampling step on the same $N = 12$ duty-cycle-prior benchmark with 30 deterministic seeds. Each QUBO-derived method uses the same accounting per seed: 96 shared trajectory evaluations to train the QUBO and 24 true post-QUBO candidate evaluations. The resulting raw package contains 150 method–seed rows; the raw method summary table is in the supplement.

The 30-seed success rates with Wilson 95% intervals are $28/30 = 0.933$ $[0.787, 0.982]$ for surrogate-QUBO simulated annealing, $28/30 = 0.933$ $[0.787, 0.982]$ for exact evaluation of the top 24 QUBO bitstrings, $15/30 = 0.500$ $[0.332, 0.668]$ for random-angle $p = 1$ QAOA, $21/30 = 0.700$ $[0.521, 0.833]$ for optimized $p = 1$ QAOA, and $29/30 = 0.967$ $[0.833, 0.994]$

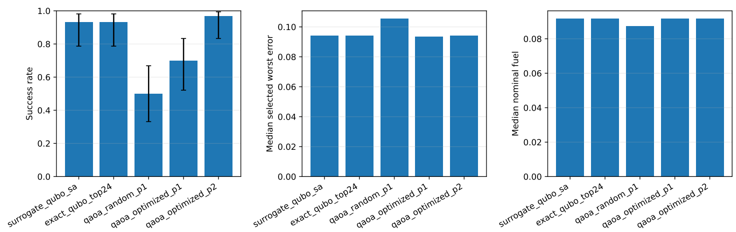

Figure 4: Thirty-seed QAOA-depth ablation summary. Optimized $p = 2$ QAOA improves over random-angle QAOA and is competitive with classical QUBO sampling, but paired tests do not support a superiority claim over surrogate-QUBO simulated annealing.

for optimized $p = 2$ QAOA. Optimized $p = 2$ is competitive with the strongest classical QUBO sampler in this narrow statevector QUBO ablation, but it is not statistically supported as superior: versus surrogate-QUBO simulated annealing, its paired success delta is only $+0.033$, the mean selected-worst-error delta is -0.00056 with bootstrap 95% confidence interval $[-0.00393, 0.00236]$, and the exact sign-test p-value is 0.4545.

The useful positive result is narrower. Relative to random-angle $p = 1$ QAOA, optimized $p = 2$ QAOA increases success by $+0.467$ and reduces the mean selected-worst error by 0.0125 with bootstrap interval $[-0.0188, -0.0060]$. The exact sign-test p-value is still not significant at conventional levels ($p = 0.2478$), so the evidence should be read as practical improvement in this ablation rather than a formal dominance claim. No quantum advantage is claimed.

6.12. Supplementary diagnostics

The supplement preserves the evidence that explains how the headline results were reached and where they fail. This includes the original catalog pilot and QUBO fit, feasibility sweeps, direct multiple-shooting and targeted catalog attempts, selected-outage hard-catalog envelope, locked-nominal and delayed-arrival hard-catalog recovery packages, bounded phase-suite and robust-margin rows, direct-collocation and Hermite–Simpson baselines, unrepaired and repaired $N = 8$ phase-shift evidence, legacy 10-seed cardinality-prior summaries, cardinality and fuel ablations, and the teacher-controlled benchmark. These artifacts support the failure-mode discussion, but the main claims do not depend on promoting them to primary results.

7. Discussion

The central result is a benchmark result, not a sampler-performance claim. Binary thrust-window schedules can be embedded in a reproducible missed-thrust initialization workflow, but the usefulness of a schedule is controlled by the continuous recovery basin. The supplementary diagnostics show both sides: sparse binary schedules can fail even when all-windows continuous refinement succeeds, while duty-cycle priors can preserve enough thrust availability for several samplers to find refinable high-duty schedules.

The 30-seed main-method package is the most relevant initializer evidence. Cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement all succeed in 30/30 seeds on the duty-cycle-prior benchmark, whereas the configured statevector QAOA row succeeds in 13/30 seeds. The paired selected-worst-error tests favor all-windows continuous refinement over every sampled method, and the stricter-threshold sensitivity check shows the sampled high-duty schedules losing pass counts before all-windows continuous does. Thus the paper supports duty-cycle-aware robust initialization as a useful benchmark mechanism, but not a claim that a binary sampler improves the continuous trajectory beyond a strong dense-control baseline.

The focused QAOA/QUBO ablation provides a more favorable but still bounded view of QAOA. Optimizing the angles and increasing simulated depth to $p = 2$ improves QAOA relative to random-angle $p = 1$ QAOA and is competitive with classical QUBO sampling in that narrow statevector study. The paired comparisons do not support superiority over surrogate-QUBO simulated annealing, and the result is not hardware evidence. For this benchmark, QAOA is best interpreted as a measured initializer family with negative and inconclusive comparisons, not as the contribution itself.

The continuous-backend ladder is useful because it separates what was optimized from what was only diagnosed. The earlier independent-midpoint Hermite–Simpson p=0.4, $a_{\max} = 0.2$ row had strong selected-branch and all-mask diagnostic errors, but it optimized only three selected outage branches. The new all-configured headroom package closes that semantic gap for the one-segment phase-shift family: all eight configured masks are optimized and evaluated. The original p=0.4 row records nominal error 0.0111 and selected/all worst error 0.0774 while reaching its configured function-evaluation cap; the polished row warm-started from the original row and its branch controls, converges, and records nominal error 0.0113 and selected/all worst error 0.0779. The replay/stress package repropagates nominal endpoint and midpoint controls at the source integration grid, refined substep grids, and direct $\pm 1\%$ acceleration scales. The new branch-control replay package separately repropagates the persisted full endpoint and midpoint branch schedules for both p=0.4 rows and records zero deltas for 16 branch rows. The bicircular phase-sweep stress package then reuses those same persisted endpoint-plus-midpoint controls under a simple circular solar-tidal perturbation and records a positive polish-row stress result over eight phases, with all nominal and branch-phase checks below the configured thresholds. The cached-Horizons-derived replay uses the same controls with interpolated Sun vectors from a representative 2026-Jan-01 JPL Horizons cache and records nominal error 0.018363195236986728 and branch worst 0.07422350563850917. The cached-Horizons Earth/Moon/Sun point-mass package records the harder negative direct replay first, then reruns independent retuning and restores retuned nominal/branch-worst errors 0.02143944130524006/0.02473065115224942 with branch pass count 8/8; the multi-epoch wrapper repeats the same point-mass retuning protocol at four fixed 2026 epochs, shows nominal direct replay fails in each epoch, and retains direct branch pass count 18/32 overall (July 8/8) plus worst retuned nominal/branch errors 0.02143944130524006/0.02473065115224942 with retuned branch pass count 32/32. The SPICE-derived replay is the next rung: it does not retune, but replays those 36 retuned controls against compact SPICE-derived Moon/Sun vector caches and records nominal/branch worst errors 0.021439441253166033/0.024730650824609506 with maximum Horizons-to-SPICE replay delta $3.2763991519857427 \times 10^{-10}$. Finally, the CasADi/IPOPT bridge exports the accepted normalized-CR3BP polish nominal and eight branch sidecars to a mature NLP backend, fixes each branch pre-recovery prefix to the refined nominal masked controls, and locally refines the source replay errors 0.011333095366088189/0.07792080291839382 to 0.009138565365046585/0.015534969964216154 with IPOPT success 9/9 and zero recorded prefix and control-bound violations. This is normalized-CR3BP continuous-backend evidence plus deterministic beyond-CR3BP stress, retuning, ephemeris-source replay, and a scoped mature-backend portability/bridge check with explicit optimizer-status and model-scope caveats, not production mission design or high-fidelity validation.

The hard-catalog tail-coast diagnostic addresses a different reviewer concern: whether any hard-catalog missed-thrust recovery case can meet the configured fixed-final-time thresholds once nominal feasibility is locked. The combined one/two-segment row does meet those thresholds, but only within a carefully scoped continuous-backend setup. The final-five-segment coast, locked nominal control, fallback starts, 25/25 optimizer-converged eligible recovery branches, and two no-recovery-variable late-tail direct evaluations are part of the result. The accepted controls replay exactly under normalized CR3BP but fail the bicircular solar-tidal stress sweep without retuning. This contrasts with the positive independent-HS simple bicircular and cached-Horizons-derived stress replays and keeps the hard-catalog result demoted to a stress limitation. A completed fixed-phase retuning batch under the same simple bicircular solar-tidal model also remains negative: at Sun phase 0°, with the original fixed target and final time and all 27 one/two-segment masks, the retuned nominal error is 0.316772, configured branch pass count is 19/27, maximum retuned branch error is 6.0299, and the strict $(0.05, 0.09)$ branch pass count is 16/27. This narrows a backend blocker and records useful recovery provenance, but it does not establish fuel-optimality, high-fidelity robustness, broader outage-family robustness, or quantum evidence.

The practical implication for astrodynamics researchers is that binary schedule benchmarks should report continuous-backend diagnostics and negative evidence as first-class artifacts. A stronger future initializer claim would need better trajectory-shaped binary encodings, mature continuous optimal-control baselines such as production collocation or sequential convex programming workflows, high-fidelity validation, and statistically significant improvements over strong classical initializers under equal total trajectory-evaluation budgets.

8. Limitations

The normalized CR3BP experiments are preliminary-design algorithm studies. They omit ephemeris dynamics, mass depletion, eclipse constraints, navigation uncertainty, thrust pointing limits beyond the acceleration norm, operational constraints, and high-fidelity force modeling. The terminal-error thresholds are screening tolerances; their dimensional interpretation is deliberately large and not comparable to flight targeting requirements.

The continuous backends are research-grade direct shooting, compact direct collocation, Hermite–Simpson, and continuation diagnostics. They are useful for comparing initializer behavior and exposing failure modes, but they are not certified flight-design tools. The new independent-midpoint Hermite–Simpson all-configured package is restricted to the normalized

CR3BP one-segment phase-shift family: the original p=0.4 row reaches its configured function-evaluation cap, while the polished p=0.4 row converges with slightly larger recorded errors. The recorded-control replay/stress table checks deterministic nominal sidecar replay only. The independent-HS branch-control replay and the focused tail-coast accepted-control replay check persisted branch-control sidecars under normalized CR3BP only; they are not optimization reruns, high-fidelity validation, broad production-solver validation/parity, fuel-optimal analysis, or quantum evidence. The independent-HS bicircular phase-sweep replay is a positive deterministic stress probe under a simple circular solar-tidal model, not SPICE/high-fidelity validation or broad production-solver validation/parity. The independent-HS cached-Horizons replay uses cached JPL Horizons Moon/Sun geometry for a representative 2026-Jan-01 simplified solar-tidal replay, but it is not SPICE propagation, high-fidelity/flight validation, flight-ready evidence, or broad production-solver validation/parity. The independent-HS cached-Horizons point-mass packages rerun local least-squares retuning after documenting persisted-control replay failure; the new multi-epoch wrapper addresses the single-epoch concern over four fixed 2026 cached-Horizons epochs, but both packages remain point-mass stress/retuning results, not SPICE propagation, full high-fidelity or flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The independent-HS SPICE-derived replay reuses the already-retuned controls against compact SPICE-derived Moon/Sun vector caches without retuning or optimization rerun; it is an Earth/Moon/Sun point-mass ephemeris-source replay, not full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence. The independent-HS CasADi/IPOPT bridge reruns a local direct-shooting refinement in a mature NLP backend for one accepted normalized-CR3BP polish case while fixing branch pre-recovery prefixes to refined nominal masked controls; it is a scoped CasADi/IPOPT mature NLP backend bridge check, not production mission design, high-fidelity or flight validation, global optimality, fuel optimality, DOI evidence, or quantum evidence. The Horizons ephemeris package is an offline force-model contrast against cached Earth/Moon/Sun geometry, not SPICE validation or ephemeris trajectory replay. The hard-catalog bicircular solar-tidal replay is a simple beyond-CR3BP stress probe and its threshold failures are an external-validity limitation, not SPICE validation. The bicircular retuned recovery package is likewise a simple fixed-phase retuning experiment under the same circular solar-tidal model; even after retuning all configured masks it fails the recorded thresholds and is not SPICE/high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, or quantum/QUBO/QAOA evidence. The claim evidence ledger clarifies selected-branch, all-mask diagnostic, all-configured-mask, branch-control replay, force-model-contrast, stress-probe, retuned-recovery, SPICE-derived ephemeris-replay, and CasADi/IPOPT bridge semantics, but it does not add high-fidelity validation. Many supplementary rows optimize selected outage branches and report unselected outage masks diagnostically; only rows that explicitly select all configured masks should be read as all-configured-mask evidence.

The hard-catalog tail-coast result is scoped to a locked-nominal continuous-backend setup with final-five-segment coast, configured one- and two-segment masks, deterministic fallback starts, and late-tail no-recovery-variable direct evaluations. The same controls do not pass the configured bicircular solar-tidal stress sweep, the retuned fixed-phase bicircular batch still fails after retuning, and the cached Horizons contrast shows distance, angular-rate, phase, and solar-tidal acceleration differences not represented by the simple circular stress model. The positive independent-HS simple bicircular phase-sweep and cached-Horizons-derived solar-tidal replays, plus the independent-HS cached-Horizons point-mass retuning packages and SPICE-derived replay, are separate one-segment phase-shift stress/retuning/replay results and do not upgrade the hard-catalog row to beyond-CR3BP robustness. The hard-catalog result therefore does not prove fuel-optimal recovery, high-fidelity robustness, certified flight design, or broader outage-family robustness.

The discrete-initializer conclusions are also bounded. The online threshold experiment is an exact simulation over finite feasible subspaces; in simulation the threshold oracle is evaluated from recorded or feedback-screening cost vectors, while query accounting separately reports cost evaluations and threshold-oracle calls. The recorded branch-refined tier covers only $M = 12$ and $M = 66$. The larger $M = 1820$ and $M = 4845$ rows are CR3BP feedback-propagation screening costs only: they use feedback propagation and terminal-error objectives, and the new targeted branch-refinement diagnostic is deliberately limited to one top screening schedule per large case. That diagnostic lowers the top schedules to selected/all-mask robust errors below 0.17, but records nominal branch errors 0.134195 and 0.115405, so both large candidates still fail the configured 0.09 nominal threshold under the default recovery cap. Therefore the large rows should not be read as branch-refined feasible trajectory evidence, high-fidelity evidence, or flight validation. The deployable classical comparators are uniform random, finite-subspace simulated annealing, cross-entropy, and genetic search under the same numeric direct-evaluation budget. The paired statistics show useful online gains in several sublinear regimes, but also show enumeration wins at full budget and genetic search can beat online on the $M = 4845$ screening case. The ranked screening row is an optimistic ceiling/diagnostic, especially for feedback-screening cases where its ranking score is closely related to the reported feedback cost; it should not be interpreted as a fair deployable baseline. The earlier predeclared Grover-style one-coast and `k10_full` experiments are retained as exact statevector diagnostics with a recorded-cost top-2 oracle; they are no longer the headline algorithm. The exploratory oracle sweep is finite and uses recorded costs to define threshold or top-k good sets, so it remains a controlled diagnostic rather than hardware evidence. The generic `qaoa_statevector` comparison remains not like-for-like because the generic baseline is still an unconstrained bitstring summary from a fitted QUBO surrogate. The auxiliary structured-QAOA packages show compatible but weaker constrained-subspace behavior. The 30-seed generic packages improve statistical rigor for the older bitstring samplers, but those generic QAOA variants are simulated with shallow depths and limited optimizer budgets and do not support QAOA superiority. Smaller supplementary packages, including teacher-controlled and legacy 10-seed results, are retained as diagnostics rather than as strong stochastic rankings.

9. Reproducibility

The repository contains Python source, configurations, source-state metadata, generated tables, figures, result metadata, cached Horizons and compact SPICE-derived vector data, PDFs, and a scoped artifact manifest. The online threshold-search package is regenerated with `py -3.11 scripts/run_online_quantum_search.py`. It writes `data/results/online_quantum_search/`, `figures/online_quantum_search/`, and `tables/online_quantum_search/`, including raw per-seed query accounting, summary CSV/JSON with Wilson intervals, paired comparison CSV/JSON, subspace cost CSVs, metadata, a success-rate figure, and the TeX tables used in the main text. The large-case branch diagnostic is regenerated with `py -3.11 scripts/run_online_branch_refinement.py`;

it reads data/results/online_quantum_search/*_subspace_costs.csv, refines one top screening schedule per default large case, and writes data/results/online_branch_refinement/online_branch_refinement.csv, data/results/online_branch_refinement/online_branch_refinement_metadata.json, and tables/online_branch_refinement/online_branch_refinement_table.tex. The older Grover-style diagnostic package is regenerated with py -3.11 scripts/run_structured_grover.py; it writes data/results/structured_grover_one_coast/, data/results/structured_grover_k10_full/, figures/structured_grover_one_coast/, figures/structured_grover_k10_full/, tables/structured_grover_one_coast/, and tables/structured_grover_k10_full/, including selection_bias_audit.csv, query_scaling.csv, and matching JSON/TeX diagnostic tables. The auxiliary structured-QAOA packages remain reproducible with py -3.11 scripts/run_structured_qaoa_one_coast.py and py -3.11 scripts/run_structured_qaoa_sensitivity.py; the k10_full package adds -group k10_full -results-dir data/results/structured_qaoa_k10_ -figures-dir figures/structured_qaoa_k10_full -tables-dir tables/structured_qaoa_k10_full. The supplement and REPRODUCIBILITY.md provide the full artifact map, including the 30-seed initializer packages, claim-evidence ledger, evidence-synthesis tables, independent-midpoint Hermite–Simpson all-configured headroom package, replay/stress sidecars, independent-HS branch-control replay, independent-HS simple bicircular phase-stress replay, independent-HS cached-Horizons solar-tidal replay, single- and multi-epoch independent-HS cached-Horizons point-mass retuning, independent-HS SPICE-derived ephemeris replay, independent-HS CasADi/IPOPT bridge artifacts, accepted tail-coast branch-control replay, cached Horizons contrast, hard-catalog bicircular stress, and negative bicircular retuning artifacts.

The short reviewer-facing check is the read-only command py -3.11 scripts/verify_submission_snapshot.py. It checks key primary artifact paths, runs scripts/write_artifact_manifest.py with -check, runs the focused reproducibility pytest subset, and runs git diff -check unless disabled by a flag. It does not regenerate artifacts, rerun trajectory optimization, rebuild PDFs, or remove LaTeX auxiliary files; -full-tests expands the pytest step to the full suite. Evidence-regeneration commands and long-run caveats remain in REPRODUCIBILITY.md.

The manifest is generated by scripts/write_artifact_manifest.py. Its scoped SHA-256 hashes and byte counts are the authoritative artifact identities. The field `git_head_at_generation` records the repository state when the manifest was produced; for a committed manifest this necessarily refers to the parent/pre-final state rather than the final commit containing the refreshed manifest. Historical long-run metadata may predate the repository or contain dirty-state records. The current submission snapshot is verified by clean git status after final commit, manifest checking, tests, and local LaTeX builds rather than by interpreting old experiment metadata as final provenance.

10. Conclusion

This paper contributes a reproducible normalized CR3BP benchmark resource and a quantum-specific constrained-search experiment for robust low-thrust cislunar initialization under missed-thrust events. The main quantum result is positive but scoped: the online threshold-amplitude-amplification protocol does not know the top-2 schedules in advance and uses only threshold comparisons against the current incumbent. On the recorded 66-state `k10_full` branch-refined subspace it improves global-optimum hit rate over uniform random, simulated annealing, cross-entropy, and genetic baselines at sublinear budgets. On CR3BP feedback-propagation screening subspaces it extends the simulation to $M = 1820$ and $M = 4845$ with strong hit-rate gains over uniform random at $4\lceil\sqrt{M}\rceil$, while also recording that a schedule-aware genetic baseline beats online on one larger screening row. The new branch-refinement diagnostic prevents that scale result from becoming an overclaim: the top large screening schedules approach/pass robust selected and all-mask diagnostics but still fail the nominal branch threshold under the default recovery cap. The ranked screening row is an intentionally optimistic ceiling/diagnostic for near-perfect classical ranking information, not a deployable baseline defeated by the online protocol. The earlier top-2 Grover rows remain useful diagnostics, but they are no longer the headline algorithm. The broader initializer evidence remains conservative: duty-cycle-aware binary schedules can help locate refinable high-duty candidates, but all-windows continuous refinement remains lower-error than the one-coast subspace optimum. Deterministic evidence-ledger, evidence-synthesis, sidecar-replay, cached Horizons force-model contrast, and solar-tidal stress layers make the validation boundaries explicit. The useful outcome is therefore an auditable benchmark with a real threshold-only constraint-preserving quantum-search simulation, larger feedback-screening scale cases, calibrated negative evidence, failure modes, and provenance, not flight-ready robustness.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

Code, generated artifacts, and artifact maps for the current draft are available in the working repository associated with this revision. The latest GitHub release/package is `v1.0.2` at commit `551eab6` (`https://github.com/Alavi1412/quantum-low-thrust/releases/tag/v1.0.2`). The current external Zenodo DOI, 10.5281/zenodo.20746480, archives the older `v1.0.1` repository/artifact snapshot at commit `af8443871bae7e0adbbe906c117ddbfe011bf207` with record URL `https://zenodo.org/records/20746480`; it should not be read as a `v1.0.2` DOI. If Zenodo later mints a `v1.0.2` DOI, the DOI fields and documentation should be updated then. The benchmark source states are derived from the NASA/JPL Three-Body Periodic Orbits API and recorded in `data/source_states.json`. The Horizons force-model contrast, independent-HS cached-Horizons solar-tidal replay, and single- and multi-epoch independent-HS cached-Horizons point-mass retuning packages run offline from committed caches in `data/cache/horizons/`, whose metadata records exact API URLs and query parameters. The SPICE-derived replay runs offline from committed compact vector caches in `data/cache/spice/`; refreshing those caches downloads NAIF kernels locally, records their SHA-256 values, and does not archive raw kernel binaries. The CasADi/IPOPT bridge runs offline from committed independent-HS sidecars with `casadi==3.7.2`. `ARCHIVAL_RELEASE.md` records both the latest GitHub release provenance and the separate older Zenodo DOI archive metadata.

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Table 1: Compact claim hierarchy for the manuscript. Pass/fail labels refer only to the configured normalized screening evidence, not to flight validation.

Claim	Primary artifacts	Status	Prohibited interpretations
Benchmark resource	Source states, configs, result CSV/JSON, generated tables/figures, cache metadata, and artifact manifest	Pass as an auditable normalized CR3BP resource	Operational targeting, flight certification, or mission-design completeness
Initializer comparison	30-seed raw results, success intervals, paired comparisons, threshold sensitivity, and QAOA/QUBO ablation tables	Negative for quantum/QAOA superiority; positive only for duty-cycle-aware refinability	Quantum advantage, hardware evidence, or sampled-method dominance over all-windows continuous refinement
Feasible-subspace amplitude amplification	Online threshold-search CSV/JSON for recorded branch-refined and CR3BP feedback-propagation screening subspaces, plus diagnostic top-2 Grover CSV/JSON, query accounting, matched uniform-random baselines, ranked ceiling diagnostics, auxiliary structured-QAOA comparisons, and generated figures/tables	Positive bounded search result: the online threshold protocol does not know top-2 schedules and scales the simulation to feedback-screening subspaces with $M = 1820$ and $M = 4845$	Hardware evidence, asymptotic speedup, quantum advantage, high-fidelity validation, dominance over all-windows continuous refinement, or superiority over a near-perfect ranked screen
Continuous-backend diagnostics	Continuation, direct collocation, Hermite–Simpson, replay/stress sidecars, and metadata packages	Mixed: several scoped CR3BP rows pass configured screens, while harder rows fail	Certified solver parity, fuel optimality, or universal missed-thrust robustness
Hard-catalog tail-coast	Tail-coast recovery CSV, accepted-control replay CSV/metadata, branch audit, and threshold audit	Scoped pass under normalized CR3BP for all 27 configured one/two masks	High-fidelity robustness, broader outage-family robustness, broad production-solver validation/parity, or quantum evidence
Force-model stress bridges	Cached Horizons contrast, independent-HS bicircular, cached-Horizons, and SPICE-derived replay CSV/metadata, independent-HS cached-Horizons point-mass retuning CSV/metadata,	Contrast-only for hard-catalog Horizons; positive simple bicircular and cached-Horizons-derived stress	Full high-fidelity SPICE trajectory propagation, ephemeris flight validation, accepted-control high-fidelity replay, broad production-solver validation/parity, or

Table 2: Cross-backend evidence synthesis generated from recorded artifacts only. Error entries are nominal / selected-worst / all-mask-worst in the normalized terminal metric. Tighter pass statuses use all-mask diagnostics conservatively; selected-branch rows remain diagnostics unless all configured masks are explicitly selected.

[illegible]	[illegible]	Role	Scope	Nom / Sel / All	Configured	Tighter or near-tight status	Practitioner lesson
[illegible]	[illegible]	sampled binary initializers versus all-windows continuous	20 deterministic seeds; selected one-segment recovery masks	counts only	see coarse threshold table	sampled methods 0/20; all-windows 20/20	Tight screening exposes margin limits of sampled high-duty schedules; the positive result is the benchmark diagnostic, not sampler superiority.
[illegible]	[illegible]	all one-segment continuous continuation from p=0.3	8/8 configured one-segment masks selected and evaluated	0.0531 / 0.0139 / 0.0139	True	0.065/0.10: True; 0.05/0.09: False	All-mask continuation has headroom at p=0.4 under 0.065/0.10 screening, but the nominal error is above 0.05.
[illegible]	[illegible]	cold all one/two-mask continuous continuation	15/15 configured one- and two-segment masks selected and evaluated	0.0632 / 0.0435 / 0.0435	True	0.065/0.10: True; 0.05/0.09: False	A broader one/two-mask phase-shift row passes 0.065/0.10 all-mask diagnostics, but not the 0.05 nominal headroom check.
[illegible]	[illegible]	compact trapezoidal selected-branch collocation	1 selected outage branch; 12 one-segment all-mask diagnostics	0.0257 / 0.0191 / 0.0602	True	0.065/0.10: True; 0.05/0.09: True	Compact collocation can be tight on this selected branch, but unselected masks remain diagnostics and should be reported.
[illegible]	[illegible]	selected-branch independent-midpoint collocation with tighter threset	2 selected outage branches; 8 one-segment all-mask diagnostics	0.0197 / 0.0108 / 0.0532	True	0.065/0.10: True; 0.05/0.09: True	Independent midpoint controls meet tight all-mask diagnostics on this phase-shift row even at amax=0.2.
[illegible]	[illegible]	all one-segment independent-midpoint collocation with tighter threset	8/8 configured one-segment masks selected and evaluated	0.0131 / 0.0774 / 0.0774	True	0.065/0.10: True; 0.05/0.09: True	The independent-midpoint HS headroom package upgrades the earlier selected-branch row to all-configured one-segment evidence at p=0.4 and amax=0.2, while retaining the max-nfev optimizer caveat.
[illegible]	[illegible]	deterministic replay of persisted endpoint-plus-midpoint controls	16 branch replay rows; 8/8 one-segment masks for original and polish rows	[illegible] / 0.0774 / 0.0779	True	0.065/0.10: replay delta only; 0.05/0.09: replay delta only	Persisted independent-HS branch controls replay exactly under normalized CR3BP; this strengthens recovery-side reproducibility without adding high-fidelity or production-solver evidence.
[illegible]	[illegible]	persisted endpoint-plus-midpoint controls replayed with simple circular solar tide	8/8 configured one-segment masks over Sun phases 0.0, 45.0, 90.0, 135.0, 180.0, 225.0, 270.0, 315.0	0.0231 / 0.0856 / 0.0856	True	0.065/0.10: True; 0.05/0.09: True	The converged independent-HS all-configured row has a positive deterministic beyond-CR3BP stress replay under the simple circular solar-tidal model; the result is useful benchmark-scoped evidence, not flight validation.
[illegible]	[illegible]	persisted endpoint-plus-midpoint controls replayed with cached JPL Horizons geometry	8/8 configured one-segment masks at representative 2026-Jan-01 epoch	0.0184 / 0.0742 / 0.0742	True	0.065/0.10: True; 0.05/0.09: True	This is stronger than the simple bicircular phase sweep because the Sun geometry comes from cached JPL Horizons vectors, but it remains a simplified stress replay, not SPICE/high-fidelity/flight validation or broad production-solver validation/parity.
[illegible]	[illegible]	independent endpoint-plus-midpoint retuning under cached-Horizons point-mass dynamics	8/8 configured one-segment masks at representative 2026-Jan-01 epoch	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	Persisted controls fail direct ephemeris point-mass replay, but independent retuning restores feasibility for nominal and all 8 branches at the representative epoch; this is not SPICE, full high-fidelity/flight validation or broad production-solver validation/parity.
[illegible]	[illegible]	independent endpoint-plus-midpoint retuning under cached-Horizons point-mass dynamics	4 representative 2026 epochs; 32 configured one-segment branch rows retuned	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	The cached-Horizons point-mass retuning result is no longer a single representative-epoch check: nominal direct replay fails in all four fixed 2026 epochs; direct branch replay passes 18/32 overall (July 8/8); retuning restores feasibility. This is still not SPICE, full high-fidelity/flight validation, broad production-solver validation/parity, fuel optimality, DOI evidence, or quantum evidence.
[illegible]	[illegible]	already-retuned endpoint-plus-midpoint controls replayed under SPICE-derived vectors	4 fixed 2026 epochs; 4 nominal and 32 branch controls replayed with no retuning	0.0214 / 0.0247 / 0.0247	True	0.065/0.10: True; 0.05/0.09: True	The multi-epoch point-mass retuned controls replay under SPICE-derived Moon/Sun vectors with sub-1e-9 normalized-error deltas relative to the cached-Horizons retuned replay. This strengthens ephemeris-source provenance but remains point-mass stress replay, not full high-fidelity/flight validation or broad production-solver validation/parity.
[illegible]	[illegible]	scoped mature NLP backend direct-shooting bridge check from accepted sidecars	1 nominal and 8 configured one-segment branch rows; branch outage segments fixed inactive; branch prefixes fixed to refined nominal masked controls; zero-variable branches=1	0.0091 / 0.0135 / 0.0255	True	0.065/0.10: True; 0.05/0.09: True	One normalized-CR3BP accepted independent-HS polish case exports to CasADi/IPOPT and locally refines under the same target/scales/branch masks while preserving source branch-recovery semantics. This is a scoped mature-backend portability/bridge check that addresses the production-solver-parity concern only in that narrow sense; it is not broad production-solver validation/parity, production mission design, high-fidelity/flight validation, global or fuel optimality, DOI evidence, or quantum evidence.
[illegible]	[illegible]	locked-nominal fixed-final-time tail-coast branch portfolio	27/27 configured one- and two-segment masks selected and evaluated	0.0230 / 0.0936 / 0.0936	True	0.05/0.10: True; 0.05/0.09: False	The hard-catalog pass is real but scoped to the tail-coast backend; it passes 0.05/0.10 and narrowly misses 0.05/0.09 because the selected/all worst error is [illegible].
[illegible]	[illegible]	bounded selected-outage independent-midpoint collocation negative diagnostic	1 selected outage branch; 15 one/two-mask all-mask diagnostics	4.2644 / 11.0843 / 11.0843	False	0.065/0.10: False; 0.05/0.09: False	The hard-catalog result is not a generic continuous-backend pass; this independent-HS selected-outage diagnostic fails badly.

Table 3: Fixed-final-time tail-coast locked-nominal hard-catalog recovery. The combined row selects/evaluates all 27 configured one- and two-segment masks in one fixed-final-time case row; separate all-single and all-two-segment rows are retained as provenance/scope rows. Branches with post-outage recovery variables are optimized. No-recovery-variable late-tail branches are direct threshold evaluations, so false branch optimizer all-success flags are accounting caveats rather than threshold failures.

Case	Policy	Outage lengths	Selected masks	Tail coast segments	Portfolio variants	Fallback eval branches	Accepted fallback branches	Eligible optimizer-converged branches	Direct no-recovery branches	Tail-coast nominal error	Selected fixed-time worst error	All-mask fixed-time diagnostic worst	Max ||u||	Bound violation	nfev	Meets thresholds
tail_coast_nominal_only_t5	none	[1]	0	5	1	0	0	0/0	0	0.0230	0.0230	27.8351	1.0000	0.0000	287	True
tail_coast_all_single_t5_portfolio	all single	[1]	14	5	2	1	1	13/13	1	0.0230	0.0230	0.0230	1.0000	0.0000	2057	True
tail_coast_all_two_segment_t5_portfolio	all configured	[2]	13	5	2	3	3	12/12	1	0.0230	0.0936	0.0936	1.0000	0.0000	3467	True
tail_coast_all_one_two_segment_t5_portfolio	all configured	[1, 2]	27	5	2	4	4	25/25	2	0.0230	0.0936	0.0936	1.0000	0.0000	5929	True

Table 4: Evidence semantics for rows commonly compared in the text. The deployable finite-subspace baselines are the fair classical comparators for the online threshold query experiment; ceiling, diagnostic, backend, and large-screening rows answer different questions.

Category	Rows	Intended reading
Fair deployable finite-subspace baselines	Uniform random sampling, finite-subspace simulated annealing, cross-entropy, genetic search	Matched-budget classical schedule searches over the same constrained cost vector.
Diagnostic/ceiling rows	Ranked screening, pre-declared top-k Grover, structured-QAOA and generic-QAOA diagnostics	Oracle, ranking, or sampler diagnostics; not deployable baseline wins or hardware evidence.
Downstream continuous controls/backends	All-windows continuous refinement; direct collocation, SCP-style continuation, CasADi/IPOPT bridge artifacts	Trajectory-optimization controls and portability checks after schedule selection, not finite-subspace quantum-search baselines.
Large feedback-screening branch diagnostic	$M = 1820$ and $M = 4845$ feedback-screening top-schedule branch refinements	Scale/query-model front-end evidence plus negative nominal-threshold diagnostic; not branch-refined feasible trajectory evidence.

Table 5: Online threshold feasible-subspace search. The fair matched-budget comparison is online threshold search versus uniform random sampling; `surrogate_ranked` is an optimistic ceiling/diagnostic ranking, not a deployable baseline claim. The online protocol uses only current-incumbent threshold comparisons and does not know top-2 states. `recorded_branch_refined` rows use recorded branch-refined costs; `feedback_screening` rows use CR3BP feedback-propagation screening costs only.

evidence_tier	case_id	M	budget_label	budget	method	successes	runs	success_rate	success_rate_wilson95_lower	success_rate_wilson95_upper	top10_percent_rate	median_best_cost	median_total_oracle_calls
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	online_threshold_AA	75	200	0.3750	0.3109	0.4439	0.5650	0.0837	4.0000
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	uniform_random	47	200	0.2350	0.1816	0.2984	0.5400	0.0837	4.0000
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	simulated_annealing	69	200	0.3450	0.2826	0.4132	0.6250	0.0837	4.0000
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	cross_entropy	55	200	0.2750	0.2178	0.3407	0.5150	0.0837	4.0000
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	genetic	55	200	0.2750	0.2178	0.3407	0.5150	0.0837	4.0000
recorded_branch_refined	one_coast_k11_full	12	sqrtM	4	surrogate_ranked	0	200	0.0000	0.0000	0.0188	0.0000	0.0892	4.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	online_threshold_AA	138	200	0.6900	0.6228	0.7500	0.7750	0.0833	8.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	uniform_random	129	200	0.6450	0.5765	0.7080	0.9150	0.0833	8.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	simulated_annealing	122	200	0.6100	0.5409	0.6749	0.8100	0.0833	8.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	cross_entropy	87	200	0.4350	0.3682	0.5043	0.7500	0.0837	8.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	genetic	88	200	0.4400	0.3730	0.5093	0.7250	0.0837	8.0000
recorded_branch_refined	one_coast_k11_full	12	2sqrtM	8	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.0833	8.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	online_threshold_AA	164	200	0.8200	0.7609	0.8671	0.8750	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	uniform_random	200	200	1.0000	0.9812	1.0000	1.0000	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	simulated_annealing	149	200	0.7450	0.6804	0.8004	0.9100	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	cross_entropy	102	200	0.5100	0.4412	0.5784	0.8200	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	genetic	102	200	0.5100	0.4412	0.5784	0.7800	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	4sqrtM	12	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	online_threshold_AA	164	200	0.8200	0.7609	0.8671	0.8750	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	uniform_random	200	200	1.0000	0.9812	1.0000	1.0000	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	simulated_annealing	149	200	0.7450	0.6804	0.8004	0.9100	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	cross_entropy	102	200	0.5100	0.4412	0.5784	0.8200	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	genetic	102	200	0.5100	0.4412	0.5784	0.7800	0.0833	12.0000
recorded_branch_refined	one_coast_k11_full	12	M	12	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.0833	12.0000
recorded_branch_refined	k10_full	66	sqrtM	9	online_threshold_AA	47	200	0.2350	0.1816	0.2984	0.7150	0.1057	9.0000
recorded_branch_refined	k10_full	66	sqrtM	9	uniform_random	14	200	0.0700	0.0422	0.1141	0.6600	0.1060	9.0000
recorded_branch_refined	k10_full	66	sqrtM	9	simulated_annealing	20	200	0.1000	0.0657	0.1494	0.6000	0.1065	9.0000
recorded_branch_refined	k10_full	66	sqrtM	9	cross_entropy	32	200	0.1600	0.1157	0.2171	0.6900	0.1059	9.0000
recorded_branch_refined	k10_full	66	sqrtM	9	genetic	32	200	0.1600	0.1157	0.2171	0.6900	0.1059	9.0000
recorded_branch_refined	k10_full	66	sqrtM	9	surrogate_ranked	0	200	0.0000	0.0000	0.0188	0.0000	0.1201	9.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	online_threshold_AA	124	200	0.6200	0.5511	0.6844	0.9250	0.1056	18.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	uniform_random	40	200	0.2000	0.1505	0.2609	0.9000	0.1057	18.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	simulated_annealing	44	200	0.2200	0.1682	0.2824	0.8700	0.1057	18.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	cross_entropy	64	200	0.3200	0.2593	0.3875	0.8700	0.1057	18.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	genetic	71	200	0.3550	0.2920	0.4235	0.9100	0.1056	18.0000
recorded_branch_refined	k10_full	66	2sqrtM	18	surrogate_ranked	0	200	0.0000	0.0000	0.0188	0.0000	0.1142	18.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	online_threshold_AA	187	200	0.9350	0.8920	0.9616	0.9900	0.1056	36.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	uniform_random	116	200	0.5800	0.5107	0.6463	1.0000	0.1056	36.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	simulated_annealing	93	200	0.4650	0.3972	0.5341	0.9900	0.1056	36.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	cross_entropy	119	200	0.5950	0.5258	0.6606	0.9700	0.1056	36.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	genetic	125	200	0.6250	0.5561	0.6891	0.9800	0.1056	36.0000
recorded_branch_refined	k10_full	66	4sqrtM	36	surrogate_ranked	0	200	0.0000	0.0000	0.0188	0.0000	0.1112	36.0000
recorded_branch_refined	k10_full	66	M	66	online_threshold_AA	199	200	0.9950	0.9722	0.9991	1.0000	0.1056	66.0000
recorded_branch_refined	k10_full	66	M	66	uniform_random	200	200	1.0000	0.9812	1.0000	1.0000	0.1056	66.0000
recorded_branch_refined	k10_full	66	M	66	simulated_annealing	133	200	0.6650	0.5970	0.7268	1.0000	0.1056	66.0000
recorded_branch_refined	k10_full	66	M	66	cross_entropy	167	200	0.8350	0.7773	0.8800	1.0000	0.1056	66.0000
recorded_branch_refined	k10_full	66	M	66	genetic	170	200	0.8500	0.7939	0.8929	1.0000	0.1056	66.0000
recorded_branch_refined	k10_full	66	M	66	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.1056	66.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	online_threshold_AA	35	200	0.1750	0.1286	0.2336	0.9950	0.7109	43.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	uniform_random	6	200	0.0300	0.0138	0.0639	1.0000	0.7236	43.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	simulated_annealing	24	200	0.1200	0.0820	0.1723	1.0000	0.7158	43.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	cross_entropy	10	200	0.0500	0.0274	0.0896	0.9950	0.7207	43.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	genetic	11	200	0.0550	0.0310	0.0958	0.9950	0.7184	43.0000
feedback_screening	feedback_N16_k12	1820	sqrtM	43	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	43.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	online_threshold_AA	89	200	0.4450	0.3778	0.5143	1.0000	0.7059	86.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	uniform_random	8	200	0.0400	0.0204	0.0769	1.0000	0.7190	86.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	simulated_annealing	52	200	0.2600	0.2041	0.3249	1.0000	0.7080	86.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	cross_entropy	28	200	0.1400	0.0987	0.1949	1.0000	0.7109	86.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	genetic	58	200	0.2900	0.2315	0.3564	1.0000	0.7070	86.0000
feedback_screening	feedback_N16_k12	1820	2sqrtM	86	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	86.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	online_threshold_AA	172	200	0.8600	0.8051	0.9013	1.0000	0.6983	172.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	uniform_random	15	200	0.0750	0.0460	0.1200	1.0000	0.7123	172.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	simulated_annealing	100	200	0.5000	0.4314	0.5686	1.0000	0.7021	172.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	cross_entropy	104	200	0.5200	0.4510	0.5882	1.0000	0.6983	172.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	genetic	163	200	0.8150	0.7554	0.8627	1.0000	0.6983	172.0000
feedback_screening	feedback_N16_k12	1820	4sqrtM	172	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	172.0000
feedback_screening	feedback_N16_k12	1820	M	1820	online_threshold_AA	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N16_k12	1820	M	1820	uniform_random	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N16_k12	1820	M	1820	simulated_annealing	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N16_k12	1820	M	1820	cross_entropy	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N16_k12	1820	M	1820	genetic	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N16_k12	1820	M	1820	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.6983	1820.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	online_threshold_AA	27	200	0.1350	0.0945	0.1893	1.0000	0.7250	70.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	uniform_random	4	200	0.0200	0.0078	0.0503	1.0000	0.7360	70.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	simulated_annealing	15	200	0.0750	0.0460	0.1200	1.0000	0.7293	70.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	cross_entropy	3	200	0.0150	0.0051	0.0432	1.0000	0.7337	70.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	genetic	13	200	0.0650	0.0384	0.1080	1.0000	0.7274	70.0000
feedback_screening	feedback_N20_k16	4845	sqrtM	70	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	70.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	online_threshold_AA	86	200	0.4300	0.3633	0.4993	1.0000	0.7219	140.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	uniform_random	7	200	0.0350	0.0171	0.0705	1.0000	0.7323	140.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	simulated_annealing	40	200	0.2000	0.1505	0.2609	1.0000	0.7248	140.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	cross_entropy	29	200	0.1450	0.1029	0.2005	1.0000	0.7250	140.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	genetic	95	200	0.4750	0.4069	0.5440	1.0000	0.7219	140.0000
feedback_screening	feedback_N20_k16	4845	2sqrtM	140	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	140.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	online_threshold_AA	175	200	0.8750	0.8220	0.9139	1.0000	0.7162	280.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	uniform_random	12	200	0.0600	0.0347	0.1019	1.0000	0.7295	280.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	simulated_annealing	66	200	0.3300	0.2686	0.3978	1.0000	0.7219	280.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	cross_entropy	120	200	0.6000	0.5308	0.6654	1.0000	0.7162	280.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	genetic	187	200	0.9350	0.8920	0.9616	1.0000	0.7162	280.0000
feedback_screening	feedback_N20_k16	4845	4sqrtM	280	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	280.0000
feedback_screening	feedback_N20_k16	4845	M	4845	online_threshold_AA	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000
feedback_screening	feedback_N20_k16	4845	M	4845	uniform_random	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000
feedback_screening	feedback_N20_k16	4845	M	4845	simulated_annealing	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000
feedback_screening	feedback_N20_k16	4845	M	4845	cross_entropy	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000
feedback_screening	feedback_N20_k16	4845	M	4845	genetic	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000
feedback_screening	feedback_N20_k16	4845	M	4845	surrogate_ranked	200	200	1.0000	0.9812	1.0000	1.0000	0.7162	4845.0000

Table 6: Paired online-threshold versus deployable classical finite-subspace baselines. Success columns are online/baseline global-optimum hits out of 200 paired seeds. Cost differences are online best cost minus baseline best cost with bootstrap 95% intervals for the mean; negative favors online. Exact sign-test values use paired best-cost differences after excluding exact ties. The surrogate-ranked diagnostic ceiling is intentionally excluded.

case_id	budget_label	baseline_method	success_online_vs_baseline	paired_success_delta_mean	mean_cost_diff_bootstrap95	sign_p
one_coast_k11_full	sqrtM	uniform	75/47	0.140000	+0.0006 [+0.0003, +0.0010]	0.8126
one_coast_k11_full	sqrtM	SA	75/69	0.030000	+0.0008 [+0.0005, +0.0011]	0.0342
one_coast_k11_full	sqrtM	CEM	75/55	0.100000	+0.0005 [+0.0001, +0.0008]	0.6998
one_coast_k11_full	sqrtM	GA	75/55	0.100000	+0.0005 [+0.0001, +0.0008]	0.6998
one_coast_k11_full	2sqrtM	uniform	138/129	0.045000	+0.0005 [+0.0002, +0.0007]	0.4884
one_coast_k11_full	2sqrtM	SA	138/122	0.080000	+0.0004 [+0.0002, +0.0006]	0.4797
one_coast_k11_full	2sqrtM	CEM	138/87	0.255000	+0.0002 [-0.0000, +0.0004]	0.0039
one_coast_k11_full	2sqrtM	GA	138/88	0.250000	+0.0002 [-0.0001, +0.0004]	0.0018
one_coast_k11_full	4sqrtM	uniform	164/200	-0.180000	+0.0004 [+0.0002, +0.0006]	2.91e-11
one_coast_k11_full	4sqrtM	SA	164/149	0.075000	+0.0002 [+0.0001, +0.0004]	0.2820
one_coast_k11_full	4sqrtM	CEM	164/102	0.310000	+0.0000 [-0.0001, +0.0003]	3.78e-07
one_coast_k11_full	4sqrtM	GA	164/102	0.310000	+0.0000 [-0.0002, +0.0002]	1.37e-06
one_coast_k11_full	M	uniform	164/200	-0.180000	+0.0004 [+0.0002, +0.0006]	2.91e-11
one_coast_k11_full	M	SA	164/149	0.075000	+0.0002 [+0.0001, +0.0004]	0.2820
one_coast_k11_full	M	CEM	164/102	0.310000	+0.0000 [-0.0001, +0.0003]	3.78e-07
one_coast_k11_full	M	GA	164/102	0.310000	+0.0000 [-0.0002, +0.0002]	1.37e-06
k10_full	sqrtM	uniform	47/14	0.165000	+0.0005 [-0.0000, +0.0010]	0.0041
k10_full	sqrtM	SA	47/20	0.135000	+0.0001 [-0.0004, +0.0007]	0.0022
k10_full	sqrtM	CEM	47/32	0.075000	+0.0006 [+0.0001, +0.0011]	1.0000
k10_full	sqrtM	GA	47/32	0.075000	+0.0006 [+0.0001, +0.0011]	1.0000
k10_full	2sqrtM	uniform	124/40	0.420000	-0.0000 [-0.0003, +0.0003]	1.05e-11
k10_full	2sqrtM	SA	124/44	0.400000	-0.0001 [-0.0004, +0.0002]	2.46e-11
k10_full	2sqrtM	CEM	124/64	0.300000	-0.0000 [-0.0003, +0.0003]	8.93e-07
k10_full	2sqrtM	GA	124/71	0.265000	+0.0000 [-0.0002, +0.0004]	1.77e-06
k10_full	4sqrtM	uniform	187/116	0.355000	-0.0000 [-0.0001, +0.0000]	1.42e-14
k10_full	4sqrtM	SA	187/93	0.470000	-0.0001 [-0.0001, -0.0000]	1.84e-22
k10_full	4sqrtM	CEM	187/119	0.340000	-0.0001 [-0.0001, -0.0000]	1.05e-13
k10_full	4sqrtM	GA	187/125	0.310000	-0.0001 [-0.0002, -0.0000]	1.39e-11
k10_full	M	uniform	199/200	-0.005000	+0.0000 [+0.0000, +0.0000]	1.0000
k10_full	M	SA	199/133	0.330000	-0.0000 [-0.0001, -0.0000]	4.68e-19
k10_full	M	CEM	199/167	0.160000	-0.0000 [-0.0000, -0.0000]	4.07e-09
k10_full	M	GA	199/170	0.145000	-0.0000 [-0.0000, -0.0000]	3.73e-09
feedback_N16_k12	sqrtM	uniform	35/6	0.145000	-0.0189 [-0.0227, -0.0152]	2.70e-20
feedback_N16_k12	sqrtM	SA	35/24	0.055000	-0.0063 [-0.0096, -0.0031]	0.0105
feedback_N16_k12	sqrtM	CEM	35/10	0.125000	-0.0158 [-0.0196, -0.0120]	6.31e-15
feedback_N16_k12	sqrtM	GA	35/11	0.120000	-0.0089 [-0.0120, -0.0058]	8.78e-09
feedback_N16_k12	2sqrtM	uniform	89/8	0.405000	-0.0165 [-0.0186, -0.0145]	3.71e-39
feedback_N16_k12	2sqrtM	SA	89/52	0.185000	-0.0045 [-0.0061, -0.0031]	4.51e-07
feedback_N16_k12	2sqrtM	CEM	89/28	0.305000	-0.0079 [-0.0092, -0.0065]	4.78e-19
feedback_N16_k12	2sqrtM	GA	89/58	0.155000	-0.0028 [-0.0039, -0.0017]	0.0002
feedback_N16_k12	4sqrtM	uniform	172/15	0.785000	-0.0141 [-0.0152, -0.0130]	8.47e-50
feedback_N16_k12	4sqrtM	SA	172/100	0.360000	-0.0041 [-0.0050, -0.0031]	1.45e-13
feedback_N16_k12	4sqrtM	CEM	172/104	0.340000	-0.0033 [-0.0040, -0.0026]	8.00e-15
feedback_N16_k12	4sqrtM	GA	172/163	0.045000	-0.0005 [-0.0011, +0.0001]	0.1770
feedback_N16_k12	M	uniform	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N16_k12	M	SA	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N16_k12	M	CEM	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N16_k12	M	GA	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N20_k16	sqrtM	uniform	27/4	0.115000	-0.0112 [-0.0128, -0.0096]	3.15e-25
feedback_N20_k16	sqrtM	SA	27/15	0.060000	-0.0039 [-0.0054, -0.0026]	0.0001
feedback_N20_k16	sqrtM	CEM	27/3	0.120000	-0.0090 [-0.0105, -0.0075]	1.84e-25
feedback_N20_k16	sqrtM	GA	27/13	0.070000	-0.0031 [-0.0042, -0.0019]	3.11e-05
feedback_N20_k16	2sqrtM	uniform	86/7	0.395000	-0.0121 [-0.0132, -0.0110]	4.10e-47
feedback_N20_k16	2sqrtM	SA	86/40	0.230000	-0.0042 [-0.0052, -0.0033]	1.89e-12
feedback_N20_k16	2sqrtM	CEM	86/29	0.285000	-0.0053 [-0.0062, -0.0044]	4.24e-21
feedback_N20_k16	2sqrtM	GA	86/95	-0.045000	+0.0001 [-0.0007, +0.0009]	1.0000
feedback_N20_k16	4sqrtM	uniform	175/12	0.815000	-0.0119 [-0.0127, -0.0110]	1.79e-52
feedback_N20_k16	4sqrtM	SA	175/66	0.545000	-0.0046 [-0.0053, -0.0040]	5.75e-27
feedback_N20_k16	4sqrtM	CEM	175/120	0.275000	-0.0021 [-0.0027, -0.0015]	1.79e-09
feedback_N20_k16	4sqrtM	GA	175/187	-0.060000	+0.0004 [+0.0000, +0.0007]	0.0652
feedback_N20_k16	M	uniform	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N20_k16	M	SA	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N20_k16	M	CEM	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000
feedback_N20_k16	M	GA	200/200	0.000000	+0.0000 [+0.0000, +0.0000]	1.0000

Table 7: Targeted branch-refinement diagnostic for the top large feedback-screening schedules selected from the online-search subspace-cost artifacts. The default run refines one schedule per case with the configured branch-recovery cap and no multistart expansion. These rows are safeguards against overclaiming: selected and all-mask robust diagnostics fall below 0.17, but both rows fail the 0.09 nominal threshold.

Case	M	Schedule	Screen obj.	Screen nom.	Screen worst	Branch nom.	Selected worst	All-mask worst	Nom. fuel	Opt. success	Pass	nfev	Outage(s)
feedback_N16_k12	1820	0000111111111111	0.698327	0.328252	0.348534	0.134195	0.153803	0.157742	0.075000	False	False	35	[9]
feedback_N20_k16	4845	00001111111111111111	0.716184	0.338560	0.355519	0.115405	0.131575	0.134621	0.080000	True	False	28	[10]

Table 8: Diagnostic one-coast feasible-subspace amplitude amplification over the 12 feasible Hamming-weight-11 schedules. The `predeclared_protocol` row is retained as a top-2-oracle ablation; `exploratory_sweep` rows are diagnostics.

method	analysis_stage	oracle	iterations	expected_selected_worst_error	probability_best_state	probability_good_set	top_schedule
predeclared_protocol_dual_candidate_warm_amplification	predeclared_protocol	top_2	2	0.0834	0.7423	0.9933	111111111110
grover_uniform_thresholdle0p0832614239111_r2	exploratory_sweep	threshold<=0.0832614239111	2	0.0833	0.9887	0.9887	111111111110
grover_uniform_top_1_r2	exploratory_sweep	top_1	2	0.0833	0.9887	0.9887	111111111110
grover_warm_thresholdle0p0836849565092_r2	exploratory_sweep	threshold<=0.0836849565092	2	0.0834	0.7423	0.9933	111111111110
grover_warm_top_2_r2	exploratory_sweep	top_2	2	0.0834	0.7423	0.9933	111111111110
grover_warm_thresholdle0p0832614239111_r11	exploratory_sweep	threshold<=0.0832614239111	11	0.0834	0.9309	0.9309	111111111110
grover_warm_top_1_r11	exploratory_sweep	top_1	11	0.0834	0.9309	0.9309	111111111110
grover_warm_thresholdle0p0832614239111_r9	exploratory_sweep	threshold<=0.0832614239111	9	0.0835	0.8783	0.8783	111111111110
grover_warm_top_1_r9	exploratory_sweep	top_1	9	0.0835	0.8783	0.8783	111111111110
grover_uniform_thresholdle0p0836849565092_r5	exploratory_sweep	threshold<=0.0836849565092	5	0.0835	0.4963	0.9925	111111011111
grover_uniform_top_2_r5	exploratory_sweep	top_2	5	0.0835	0.4963	0.9925	111111011111
grover_warm_thresholdle0p0836849565092_r7	exploratory_sweep	threshold<=0.0836849565092	7	0.0835	0.7029	0.9405	111111111110

Table 9: Exploratory best warm-started one-coast Grover row versus feasible-subspace and QAOA comparators. This table is retained as a diagnostic; the predeclared protocol audit is Table 10.

comparison	grover_method	grover_expected_selected_worst_error	baseline_method	baseline_expected_selected_worst_error	delta_selected_worst_error	grover_probability_best_state	baseline_probability_best_state
best_grover_warm_minus_uniform_feasible	grover_warm_thresholdle0p0836849565092_r2	0.0834	uniform_feasible	0.0875	-0.0041	0.7423	0.0833
best_grover_warm_minus_warm_start_only	grover_warm_thresholdle0p0836849565092_r2	0.0834	warm_start_only	0.0843	-0.0009	0.7423	0.4774
best_grover_warm_minus_matched_uniform_grover	grover_warm_thresholdle0p0836849565092_r2	0.0834	grover_uniform_thresholdle0p0836849565092_r2	0.0847	-0.0013	0.7423	0.3714
best_grover_warm_minus_best_structured_qaoa	grover_warm_thresholdle0p0836849565092_r2	0.0834	warm_start_xy_qaoa_p4	0.0835	-0.0001	0.7423	0.6531
best_grover_warm_minus_generic_qaoa_median	grover_warm_thresholdle0p0836849565092_r2	0.0834	qaoa_statevector_30seed_median	0.1208	-0.0374	0.7423	NaN

Table 10: One-coast predeclared-protocol selection-bias audit. Deltas are comparator minus the predeclared expected selected-worst error, so positive values favor the predeclared row. The generic QAOA median is descriptive and not like-for-like.

row_type	method	expected_selected_worst_error	delta_vs_predeclared	probability_best_state	predeclared_equals_best_warm_sweep
predeclared_protocol	predeclared_protocol_dual_candidate_warm_amplification	0.0834	0.0000	0.7423	True
best_warm_start_sweep_row	grover_warm_thresholdle0p0836849565092_r2	0.0834	0.0000	0.7423	True
uniform_feasible	uniform_feasible	0.0875	0.0041	0.0833	True
warm_start_only	warm_start_only	0.0843	0.0009	0.4774	True
matched_uniform_grover_same_oracle_r	grover_uniform_top_2_r2	0.0847	0.0013	0.3714	True
best_structured_qaoa	warm_start_xy_qaoa_p4	0.0835	0.0001	0.6531	True
generic_qaoa_median_descriptive_only	qaoa_statevector_30seed_median_descriptive_only	0.1208	0.0374	NaN	True

Table 11: One-coast query and scaling accounting for the predeclared protocol. Query-scaling notes are theoretical context for amplitude amplification and minimum finding, not hardware or advantage evidence.

protocol	M	marked_count	initial_good_probability	selected_iterations	oracle_calls_per_sample	probability_best_state	expected_selected_worst_error	classical_exhaustive_M	warm_start_random_samples_to_match_p_best
dual_candidate_warm_amplification	12	2	0.6388	2	2	0.7423	0.0834	12	3.0000

Table 12: Diagnostic `k10_full` feasible-subspace amplitude amplification over the 66 feasible Hamming-weight-10 schedules. The `predeclared_protocol` row is retained as a top-2-oracle ablation; `exploratory_sweep` rows are diagnostics.

method	analysis_stage	oracle	iterations	expected_selected_worst_error	probability_best_state	probability_good_set	top_schedule
predeclared_protocol_dual_candidate_warm_amplification	predeclared_protocol	top_2	1	0.1056	0.5142	0.9963	111111111100
grover_uniform_thresholdle0p105574892545_r6	exploratory_sweep	threshold<=0.105574892545	6	0.1056	0.9989	0.9989	111111111100
grover_uniform_top_1_r6	exploratory_sweep	top_1	6	0.1056	0.9989	0.9989	111111111100
grover_uniform_top_2_r4	exploratory_sweep	top_2	4	0.1056	0.5000	1.0000	111111111001
grover_warm_top_2_r1	exploratory_sweep	top_2	1	0.1056	0.5142	0.9963	111111111100
grover_warm_thresholdle0p105574892545_r6	exploratory_sweep	threshold<=0.105574892545	6	0.1056	0.9877	0.9877	111111111100
grover_warm_top_1_r6	exploratory_sweep	top_1	6	0.1056	0.9877	0.9877	111111111100
grover_warm_top_3_r6	exploratory_sweep	top_3	6	0.1056	0.3726	1.0000	111111111100
grover_uniform_top_3_r3	exploratory_sweep	top_3	3	0.1057	0.3318	0.9955	111111111001
grover_warm_thresholdle0p105574892545_r2	exploratory_sweep	threshold<=0.105574892545	2	0.1057	0.9610	0.9610	111111111100
grover_warm_top_1_r2	exploratory_sweep	top_1	2	0.1057	0.9610	0.9610	111111111100
grover_warm_top_5_r11	exploratory_sweep	top_5	11	0.1057	0.2563	0.9980	111111111100

Table 13: Exploratory best warm-started `k10_full` Grover row versus feasible-subspace and QAOA comparators. This table is retained as a diagnostic; the predeclared protocol audit is Table 14.

comparison	grover_method	grover_expected_selected_worst_error	baseline_method	baseline_expected_selected_worst_error	delta_selected_worst_error	grover_probability_best_state	baseline_probability_best_state
best_grover_warm_minus_uniform_feasible	grover_warm_top_2_r1	0.1056	uniform_feasible	0.1138	-0.0082	0.5142	0.0152
best_grover_warm_minus_warm_start_only	grover_warm_top_2_r1	0.1056	warm_start_only	0.1087	-0.0031	0.5142	0.1201
best_grover_warm_minus_matched_uniform_grover	grover_warm_top_2_r1	0.1056	grover_uniform_top_2_r1	0.1119	-0.0063	0.5142	0.1256
best_grover_warm_minus_best_structured_qaoa	grover_warm_top_2_r1	0.1056	structured_inverse_complete_p4_r8	0.1062	-0.0006	0.5142	0.1800
best_grover_warm_minus_generic_qaoa_median	grover_warm_top_2_r1	0.1056	qaoa_statevector_30seed_median	0.1208	-0.0152	0.5142	NaN

Table 14: `k10_full` predeclared-protocol selection-bias audit. Deltas are comparator minus the predeclared expected selected-worst error, so positive values favor the predeclared row. The generic QAOA median is descriptive and not like-for-like.

row_type	method	expected_selected_worst_error	delta_vs_predeclared	probability_best_state	predeclared_equals_best_warm_sweep
predeclared_protocol	predeclared_protocol_dual_candidate_warm_amplification	0.1056	0.0000	0.5142	True
best_warm_start_sweep_row	grover_warm_top_2_r1	0.1056	0.0000	0.5142	True
uniform_feasible	uniform_feasible	0.1138	0.0082	0.0152	True
warm_start_only	warm_start_only	0.1087	0.0031	0.1201	True
matched_uniform_grover_same_oracle_r	grover_uniform_top_2_r1	0.1119	0.0063	0.1256	True
best_structured_qaoa	structured_inverse_complete_p4_r8	0.1062	0.0006	0.1800	True
generic_qaoa_median_descriptive_only	qaoa_statevector_30seed_median_descriptive_only	0.1208	0.0152	NaN	True

Table 15: `k10_full` query and scaling accounting for the predeclared protocol. Query-scaling notes are theoretical context for amplitude amplification and minimum finding, not hardware or advantage evidence.

protocol	M	marked_count	initial_good_probability	selected_iterations	oracle_calls_per_sample	probability_best_state	expected_selected_worst_error	classical_exhaustive_M	warm_start_random_samples_to_match_p_best
dual_candidate_warm_amplification	66	2	0.2328	1	1	0.5142	0.1056	66	6.0000

Table 16: Thirty-seed $N = 12$ duty-cycle-prior phase-shift method comparison. Values summarize 210 method–seed rows. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0928	0.1188	0.1277	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.3667
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0948	0.1208	0.1299	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4333
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 17: Thirty-seed main-method statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method. Very small sign-test values are rendered in scientific notation rather than rounded to zero.

method	success	wilson_95_ci	delta_success_vs_all_windows_continuous	mean_error_delta_vs_all_windows_continuous	sign_p_vs_all_windows_continuous	delta_success_vs_surrogate_qubo_sa	mean_error_delta_vs_surrogate_qubo_sa	sign_p_vs_surrogate_qubo_sa
random	11/30	[0.22, 0.54]	-0.63	+0.0595 [+0.0533, +0.0659]	1.86e-09	-0.63	+0.0264 [+0.0201, +0.0328]	5.95e-05
cross_entropy	30/30	[0.89, 1.00]	+0.00	+0.0333 [+0.0327, +0.0337]	1.86e-09	+0.00	+0.0002 [-0.0006, +0.0009]	0.6291
genetic	30/30	[0.89, 1.00]	+0.00	+0.0325 [+0.0318, +0.0331]	1.86e-09	+0.00	-0.0006 [-0.0015, +0.0002]	0.3323
true_sa	30/30	[0.89, 1.00]	+0.00	+0.0334 [+0.0329, +0.0338]	1.86e-09	+0.00	+0.0003 [-0.0004, +0.0010]	0.8238
surrogate_qubo_sa	30/30	[0.89, 1.00]	+0.00	+0.0331 [+0.0326, +0.0335]	1.86e-09			
qaoa_statevector	13/30	[0.27, 0.61]	-0.57	+0.0484 [+0.0422, +0.0544]	1.86e-09	-0.57	+0.0153 [+0.0090, +0.0214]	0.1849
all_windows_continuous	30/30	[0.89, 1.00]				+0.00	-0.0331 [-0.0335, -0.0326]	1.86e-09

Table 18: Threshold-sensitivity counts recomputed from the recorded 30-seed raw CSV. A row passes when both `refined_nominal_error` and `refined_selected_worst_error` are below the listed thresholds. No trajectory optimization is rerun.

Threshold	(e_n, e_s)	Random	CEM	Genetic	True SA	QUBO SA	QAOA	All-windows
Configured	(0.090, 0.170)	11/30	30/30	30/30	30/30	30/30	13/30	30/30
Stricter	(0.075, 0.120)	4/30	30/30	30/30	30/30	30/30	13/30	30/30
Stringent	(0.065, 0.100)	3/30	2/30	4/30	1/30	2/30	8/30	30/30
Tight	(0.050, 0.090)	0/30	0/30	0/30	0/30	0/30	0/30	30/30

Table 19: Thirty-seed QAOA/QUBO statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method.

method	success	wilson_95_ci	delta_success_vs_sa	mean_error_delta_vs_sa	sign_p_vs_sa	delta_success_vs_random_qaoa	mean_error_delta_vs_random_qaoa	sign_p_vs_random_qaoa
surrogate_qubo_sa	28/30	[0.79, 0.98]						
exact_qubo_top24	28/30	[0.79, 0.98]	+0.00	+0.0006 [-0.0026, +0.0039]	0.3323			
qaoa_random_p1	15/30	[0.33, 0.67]	-0.43	+0.0119 [+0.0053, +0.0184]	0.5716			
qaoa_optimized_p1	21/30	[0.52, 0.83]	-0.23	+0.0048 [-0.0012, +0.0109]	1.0000	+0.20	-0.0072 [-0.0141, -0.0001]	0.3616
qaoa_optimized_p2	29/30	[0.83, 0.99]	+0.03	-0.0006 [-0.0039, +0.0024]	0.4545	+0.47	-0.0125 [-0.0188, -0.0060]	0.2478

Table 20: Compact scientific lessons from the current evidence package.

Lesson	Evidence and boundary
Dense-control baseline	All-windows continuous refinement is the necessary dense-control baseline. It remains lower-error in paired selected-worst comparisons and still passes 30/30 seeds at the tight $(0.050, 0.090)$ threshold where every sampled method is 0/30.
Evidence semantics	Selected-branch evidence, all-mask diagnostics, and all-configured-mask evidence must be separated. A row is not all-configured evidence unless it explicitly optimizes/evaluates every configured mask in that scope.
CR3BP headroom	The all-configured independent-midpoint Hermite–Simpson package gives phase-shift CR3BP headroom for eight one-segment masks: the original p=0.4 row has nominal error 0.0111 and all-configured worst error 0.0774 but retains the `max_nfev`/`optimizer_success=False` caveat; the polished row converges with nominal error 0.0113 and worst error 0.0779. Persisted branch endpoint/midpoint controls replay exactly under normalized CR3BP. A scoped CasADi/IPOPT mature NLP backend bridge check for one normalized-CR3BP accepted case refines the accepted polish nominal/branch-worst errors to 0.00914/0.01553 with IPOPT success 9/9 while preserving branch-recovery semantics, but it is a local bridge check rather than production mission design or broad production-solver validation.
External validity	The converged independent-HS polish row passes a simple bicircular phase sweep with maximum nominal/branch errors 0.02214/0.08557 and a cached-Horizons-derived representative-epoch replay with errors 0.01836/0.07422 and branch pass count 8/8. Cached-Horizons Earth/Moon/Sun point-mass replay records persisted-control failure (0.38126 nominal); independent retuning restores 0.02144 nominal and 0.02473 branch-worst errors for all 8 branches at the January epoch, and the new four-epoch wrapper repeats the stress/retuning check across Jan/Apr/Jul/Oct 2026, where nominal direct replay fails in 4/4 epochs, direct branch pass count is 18/32 overall (July 8/8), and retuned branch pass count is 32/32. A SPICE-derived replay of those retuned controls keeps branch pass count 32/32 with maximum